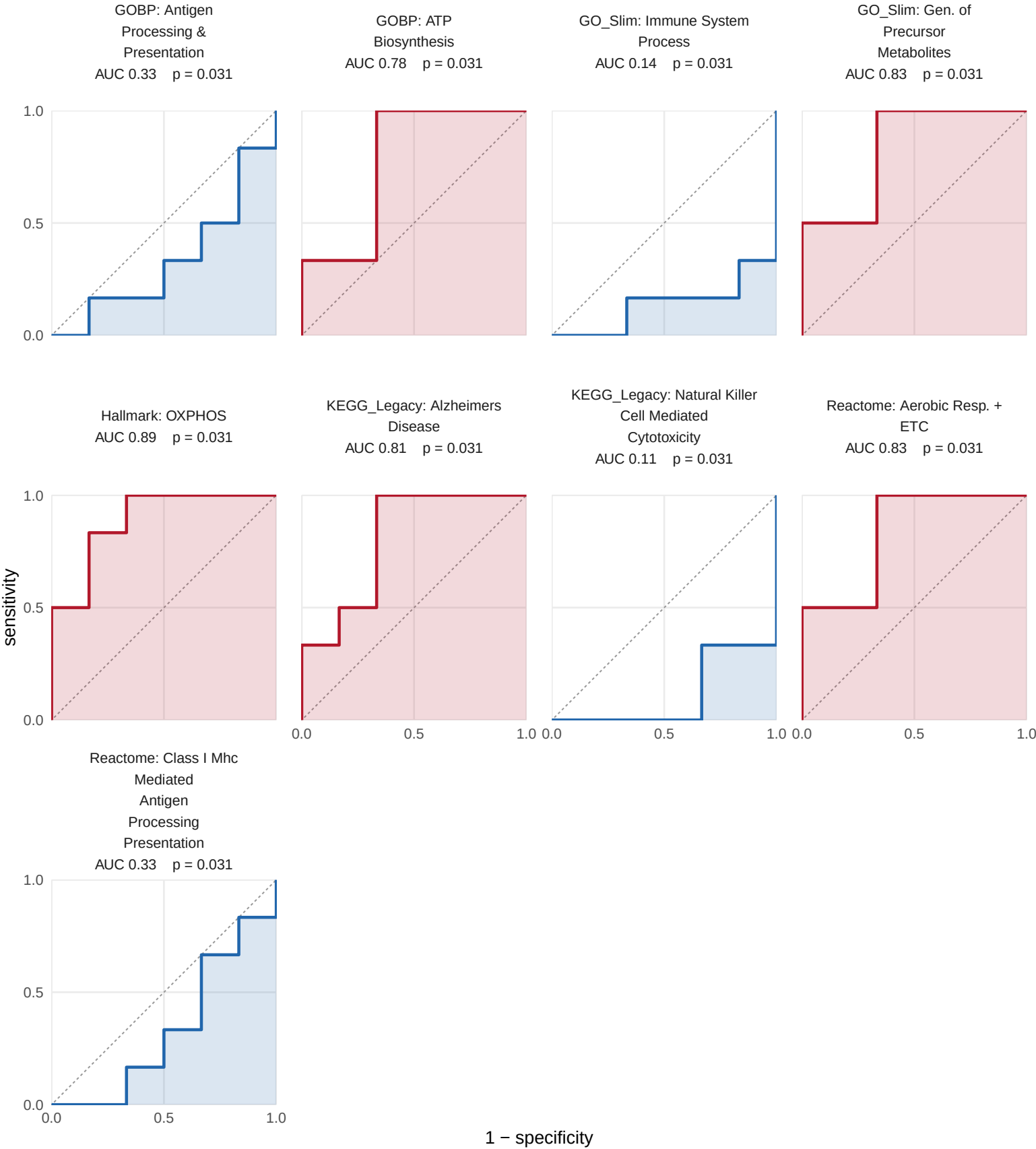


Training, HR (T1 to T2)

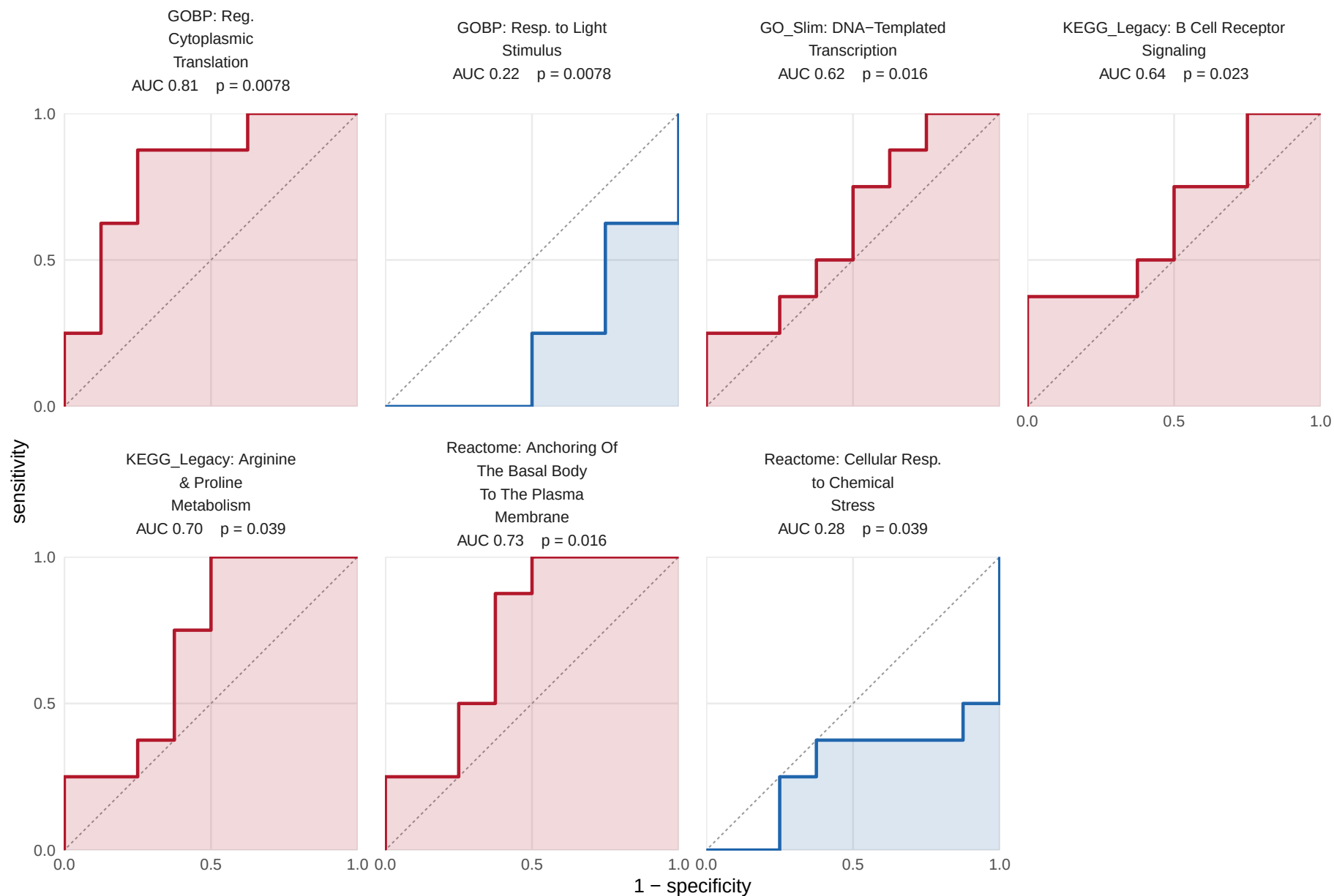
singscore per set, 6 paired subjects, AUC with direction fixed



Strongest 2 sets per collection of 50 reaching nominal p. Red separates higher in T2, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon signed-rank test. N p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Training, LR (T1 to T2)

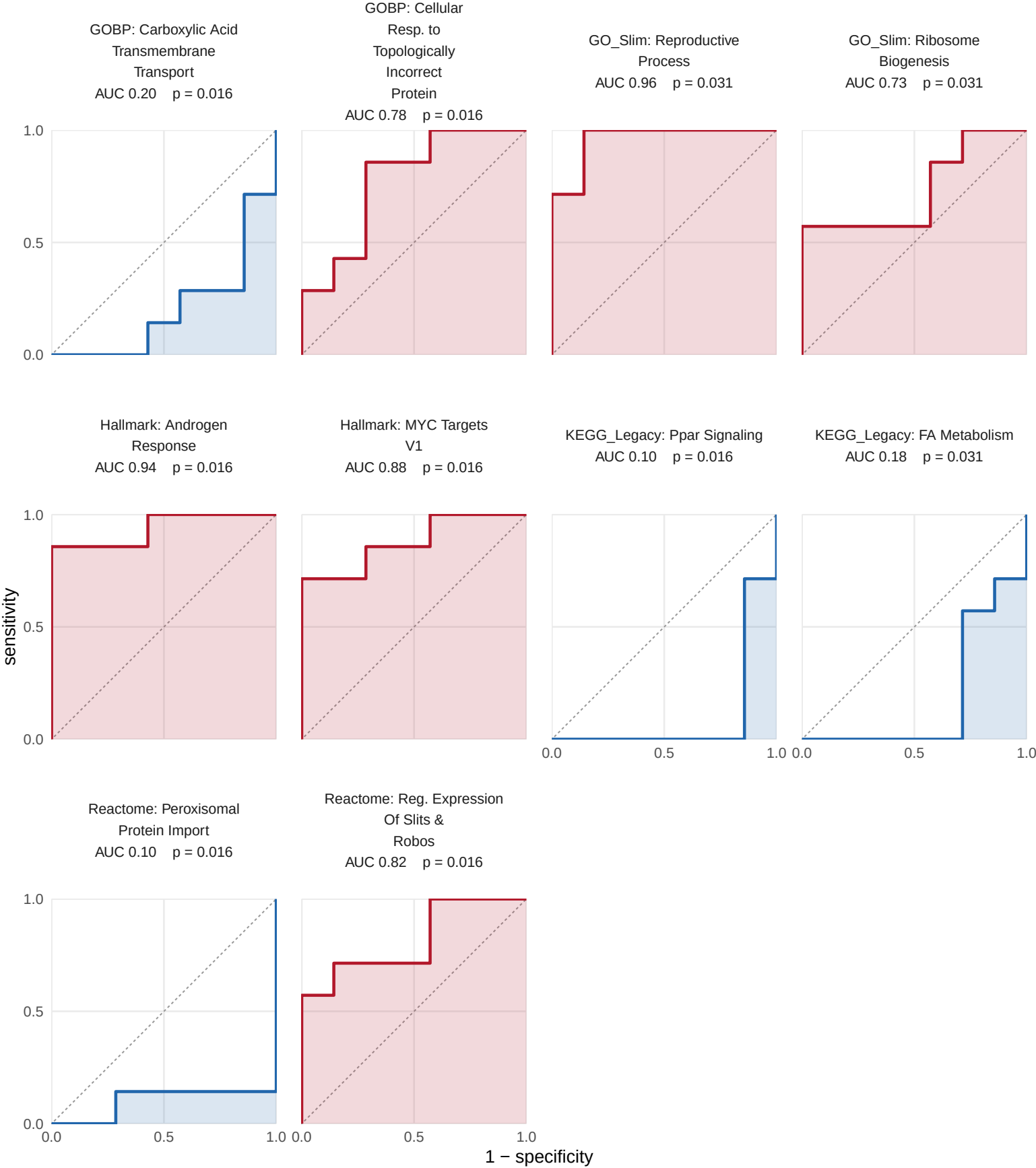
score per set, 8 paired subjects, AUC with direction fixed



Strongest 2 sets per collection of 30 reaching nominal p. Red separates higher in T2, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon signed-rank test. Nominal p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Acute bout, HR (T2 to T3)

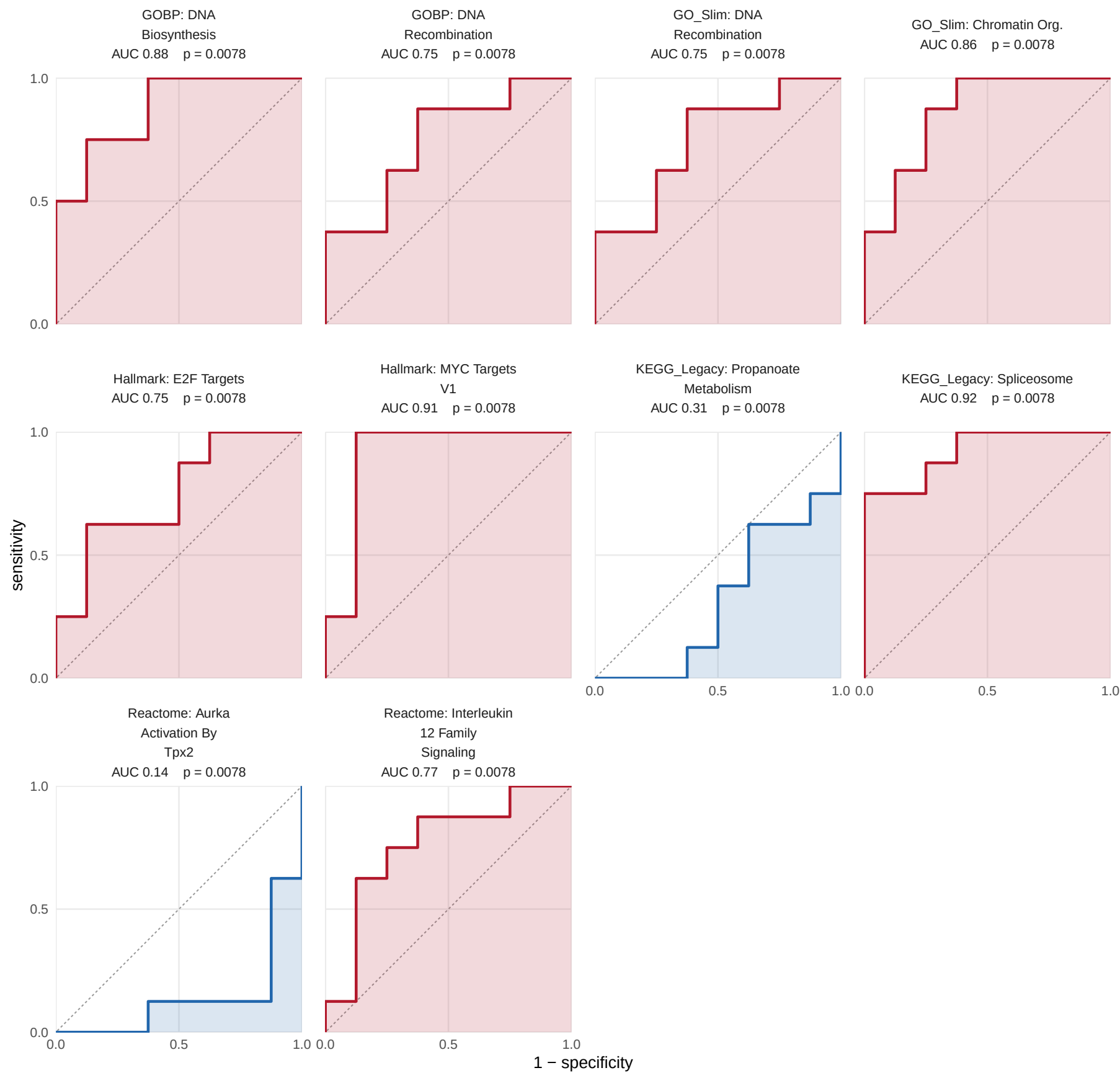
singscore per set, 7 paired subjects, AUC with direction fixed



Strongest 2 sets per collection of 113 reaching nominal p. Red separates higher in T3, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon signed-rank test. p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Acute bout, LR (T2 to T3)

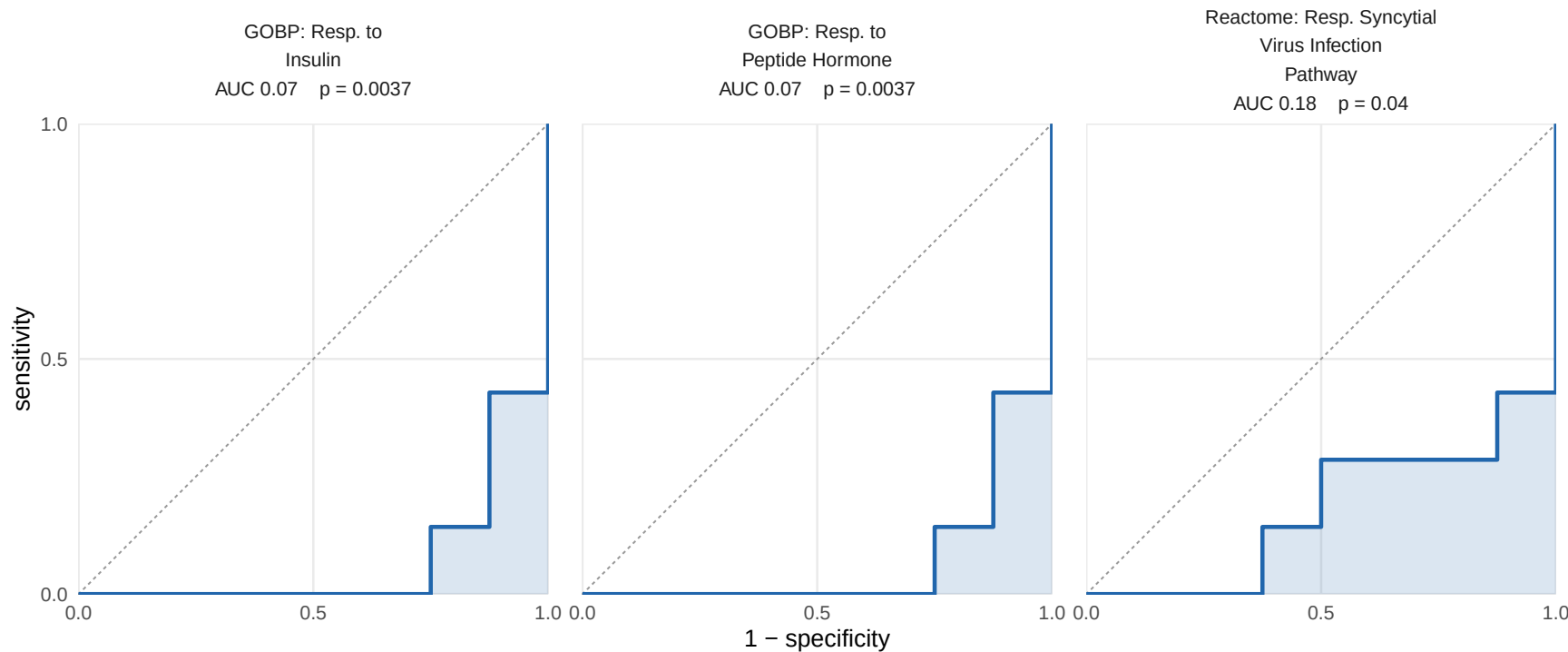
singscore per set, 8 paired subjects, AUC with direction fixed



Strongest 2 sets per collection of 153 reaching nominal p. Red separates higher in T3, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon signed-rank test. Nominal p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

HR vs LR at T2

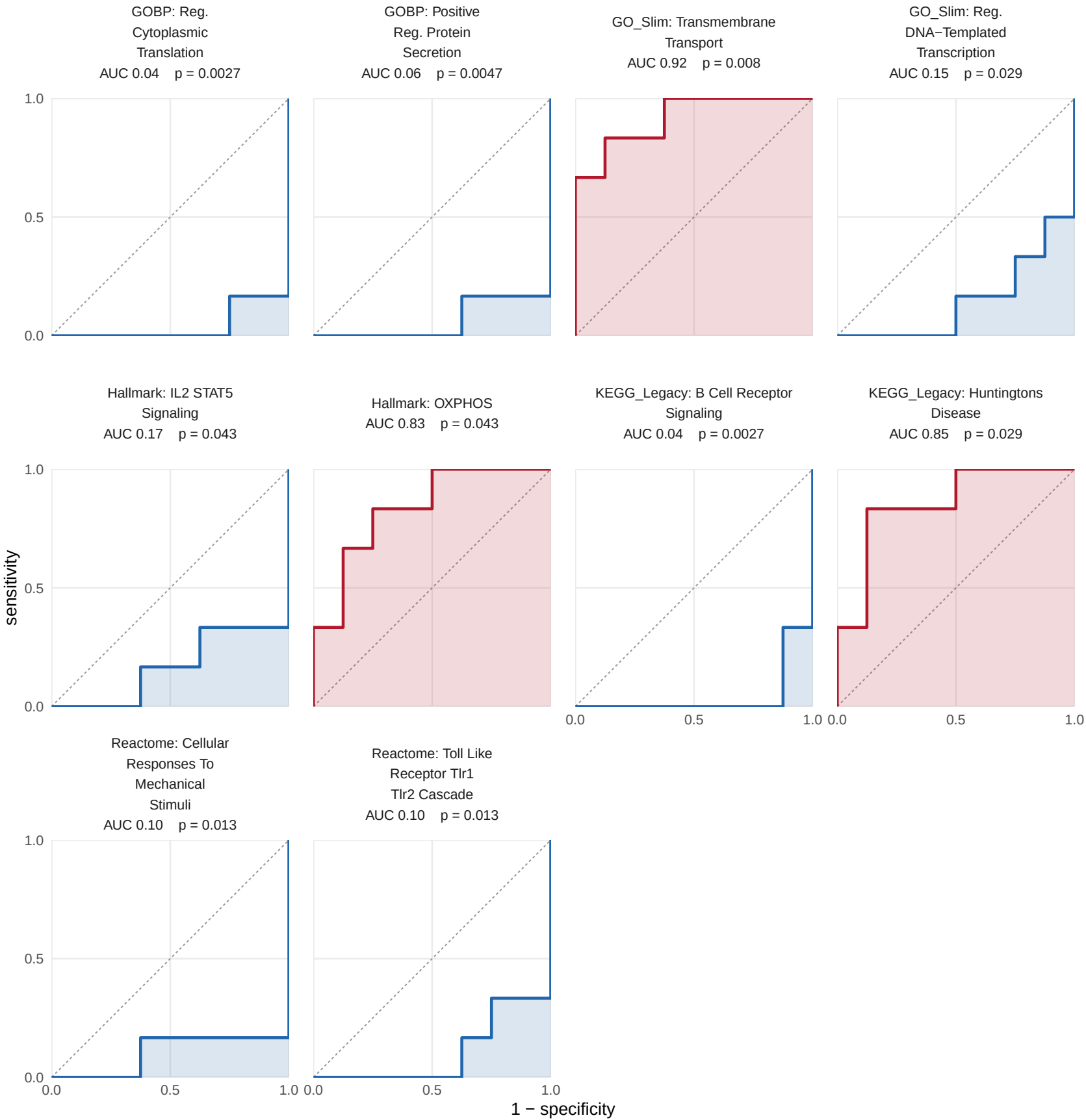
singscore per set, 7 subjects, AUC with direction fixed



Strongest 2 sets per collection of 14 reaching nominal p. Red separates higher in HR, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon rank-sum test. Nominal p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

HR vs LR, training change

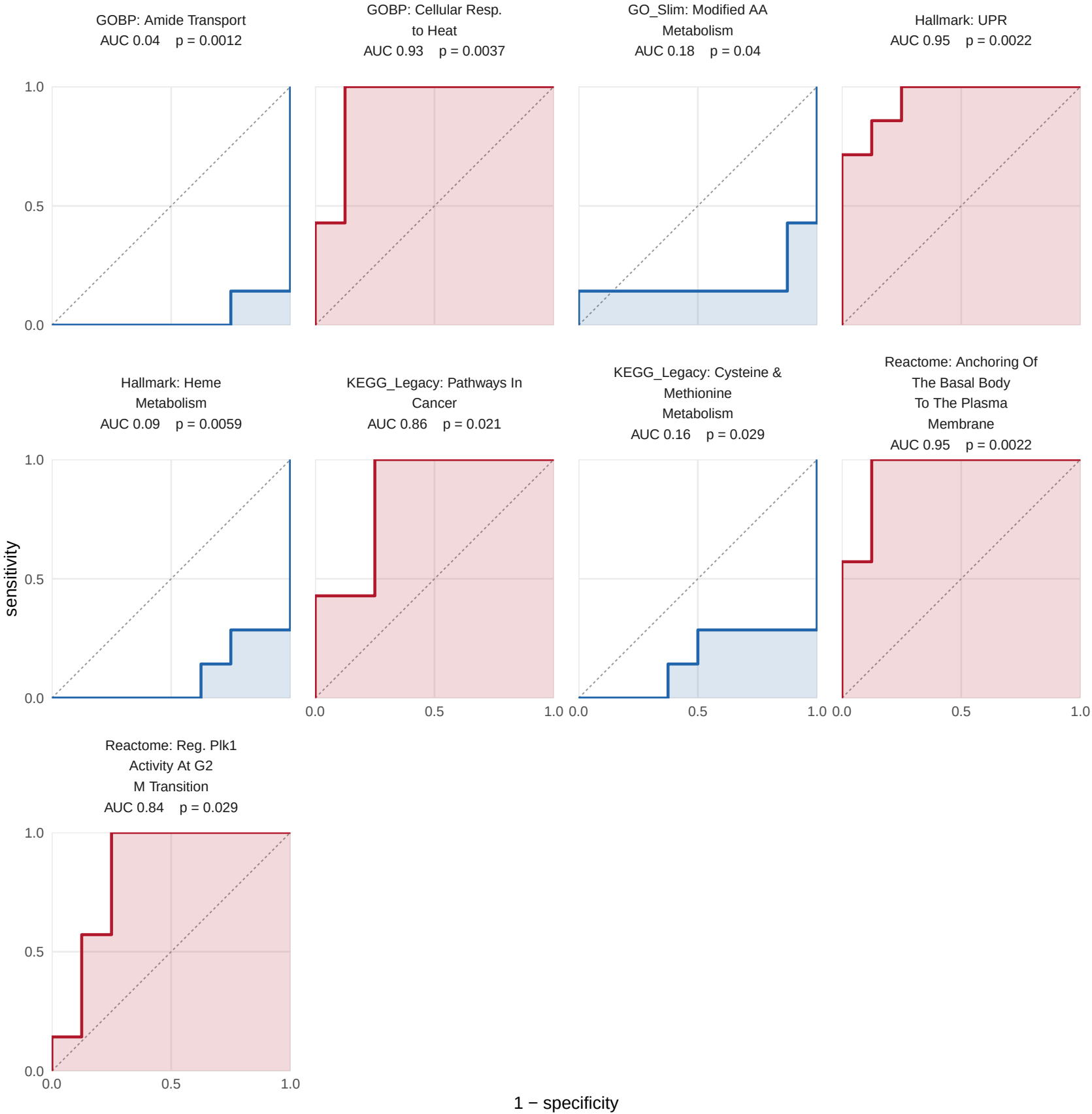
singscore per set, 6 subjects, AUC with direction fixed



Strongest 2 sets per collection of 37 reaching nominal p. Red separates higher in HR, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon rank-sum test. Nominal p uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

HR vs LR, acute change

singscore per set, 7 subjects, AUC with direction fixed

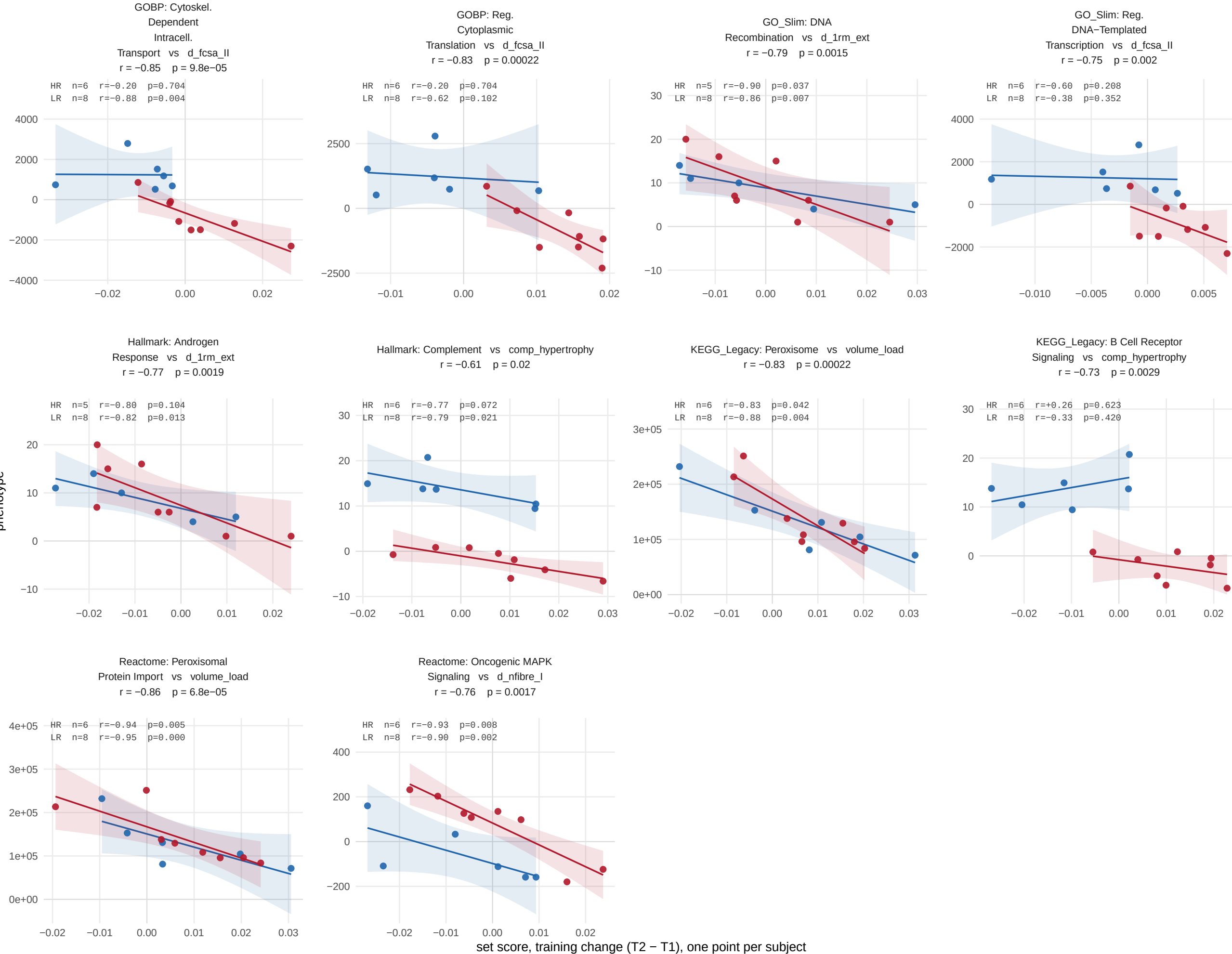


Strongest 2 sets per collection of 56 reaching nominal p. Red separates higher in HR, blue lower. Shading is area under the curve, dashed line chance. p from the Wilcoxon rank-sum test. Nominal p uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Set score against phenotype, training change (T2 – T1)

Spearman, subjects pooled across arms

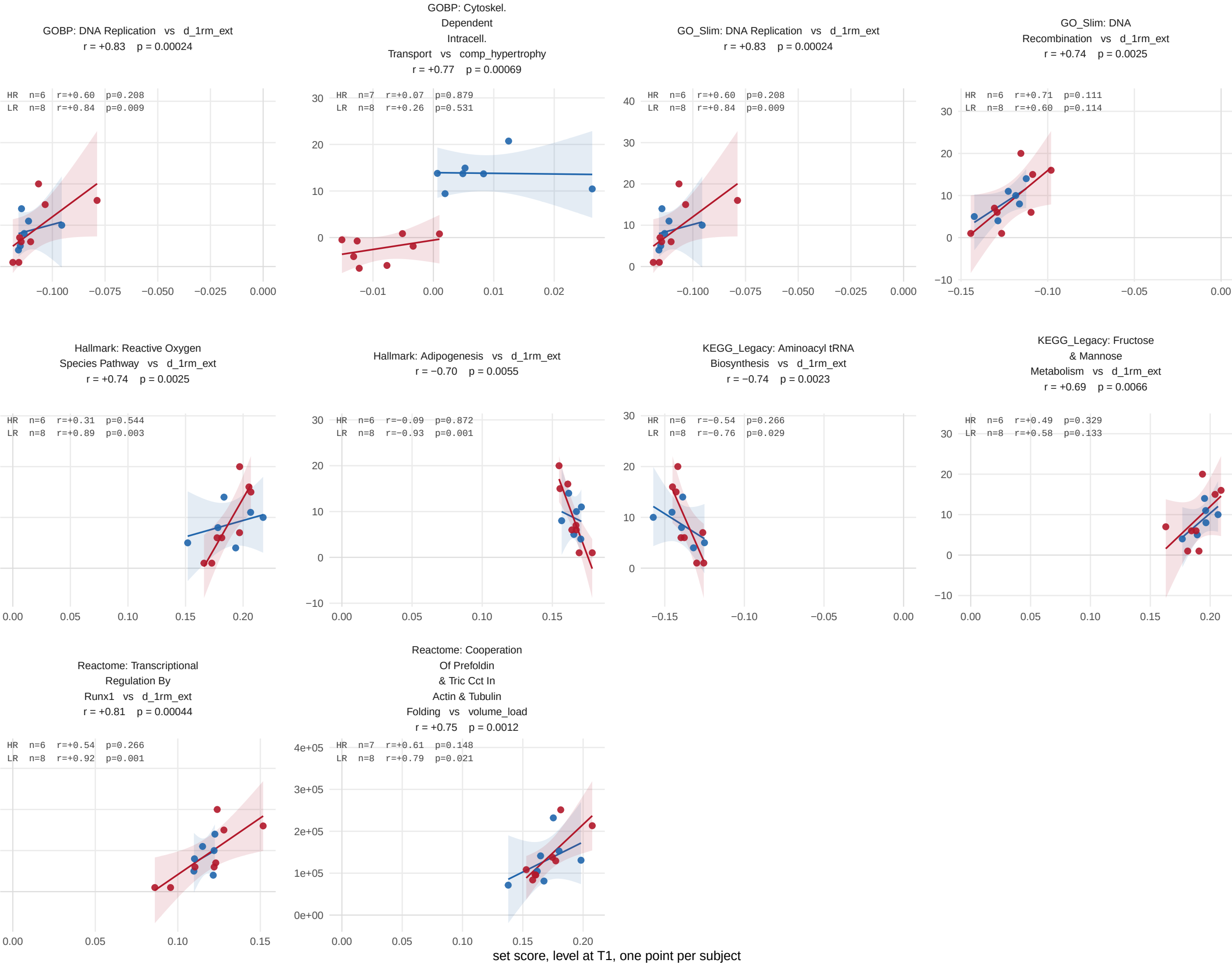
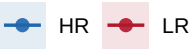
HR LR



Strongest 2 set-outcome pairs per collection of 518 reaching nominal p. Lines are fitted within each arm; the header r is the pooled correlation that selected the panel. Nominal p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Set score against phenotype, level at T1

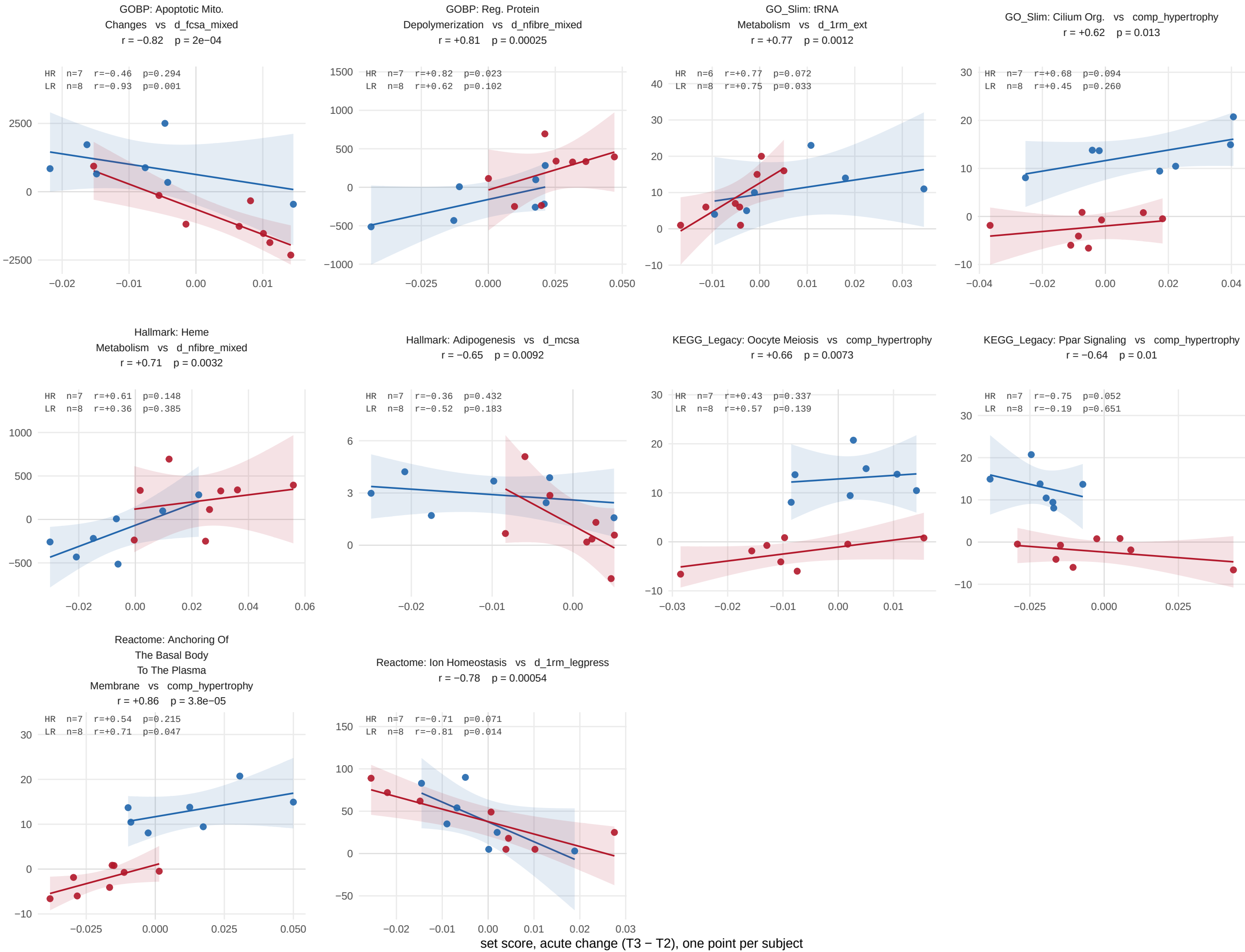
Spearman, subjects pooled across arms



Strongest 2 set-outcome pairs per collection of 576 reaching nominal p. Lines are fitted within each arm; the header r is the pooled correlation that selected the panel. Nominal p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Set score against phenotype, acute change (T3 – T2)

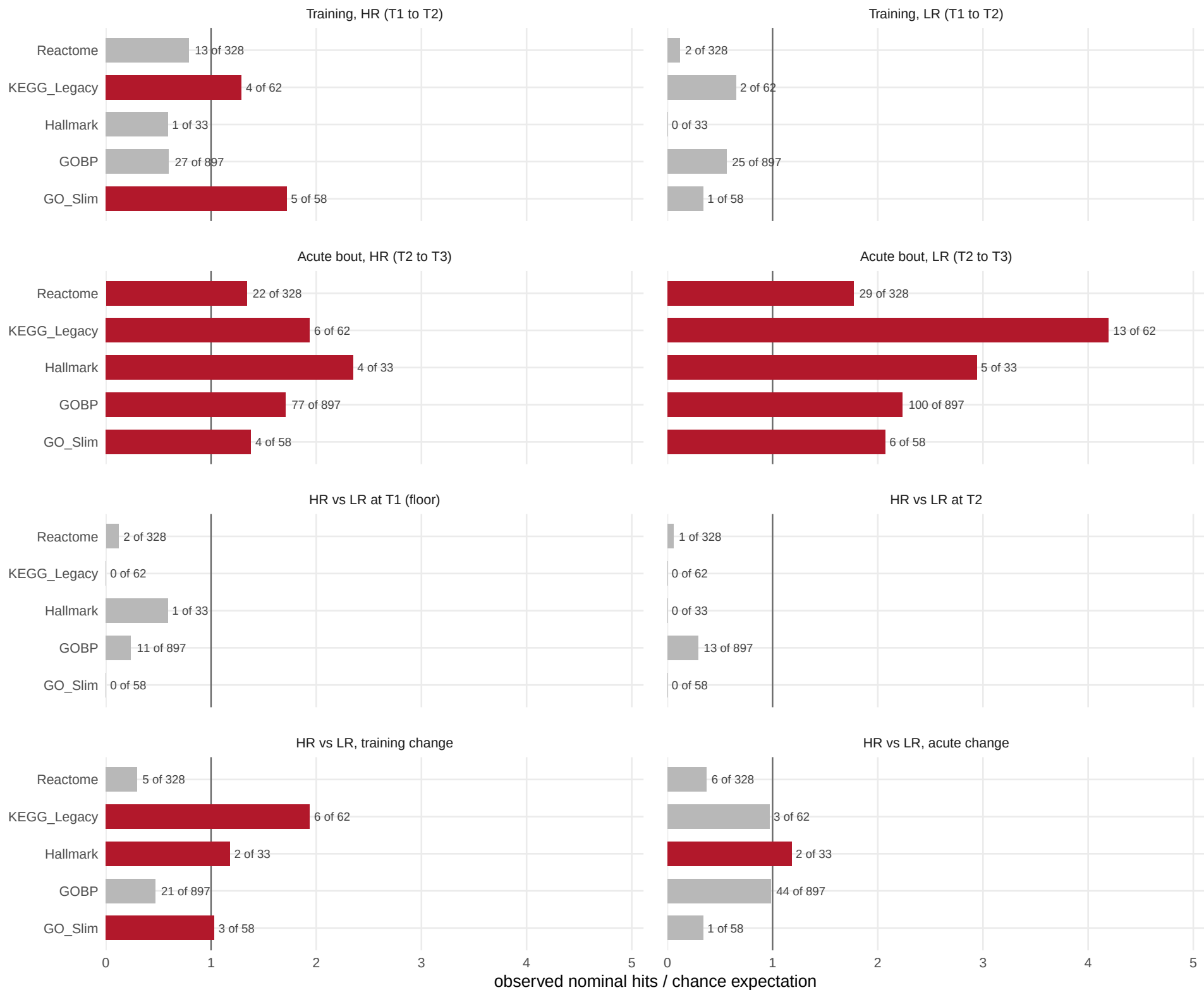
Spearman, subjects pooled across arms



Strongest 2 set-outcome pairs per collection of 511 reaching nominal p. Lines are fitted within each arm; the header r is the pooled correlation that selected the panel. Nominal p, uncorrected. Each collection is read against its own expectation: GOBP 44.9 of 897, GO_Slim 2.9 of 58, Hallmark 1.7 of 33, KEGG_Legacy 3.1 of 62, Reactome 16.4 of 328.

Nominal hits relative to chance, by collection

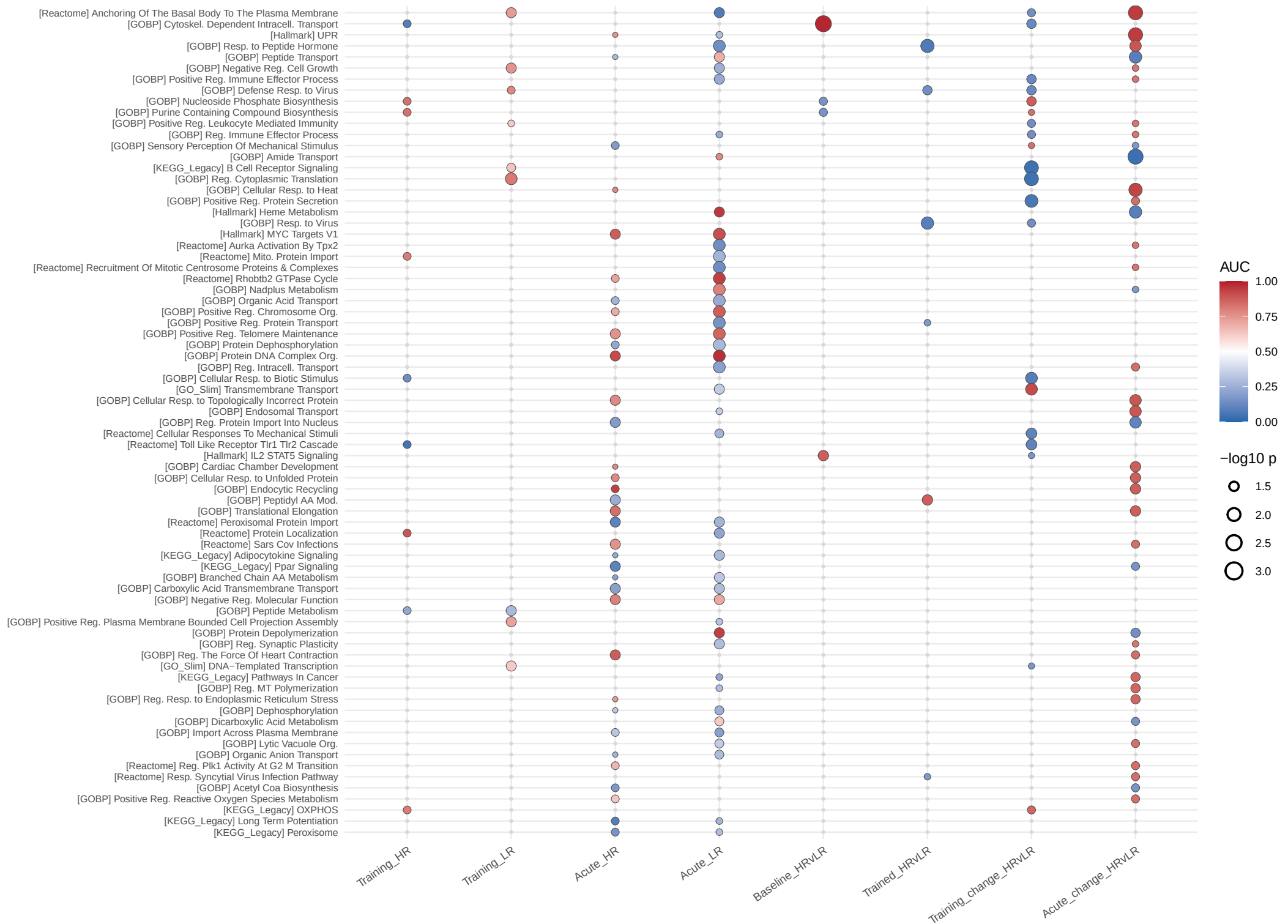
Wilcoxon per set, 5 collections, uncorrected p



Bar length is observed nominal hits divided by the count that collection returns under the null. Red clears 1, grey does not. Labels give observed of tested. Table: c_data/05_classify_and_associate_sets.xlsx, chance_expectation sheet.

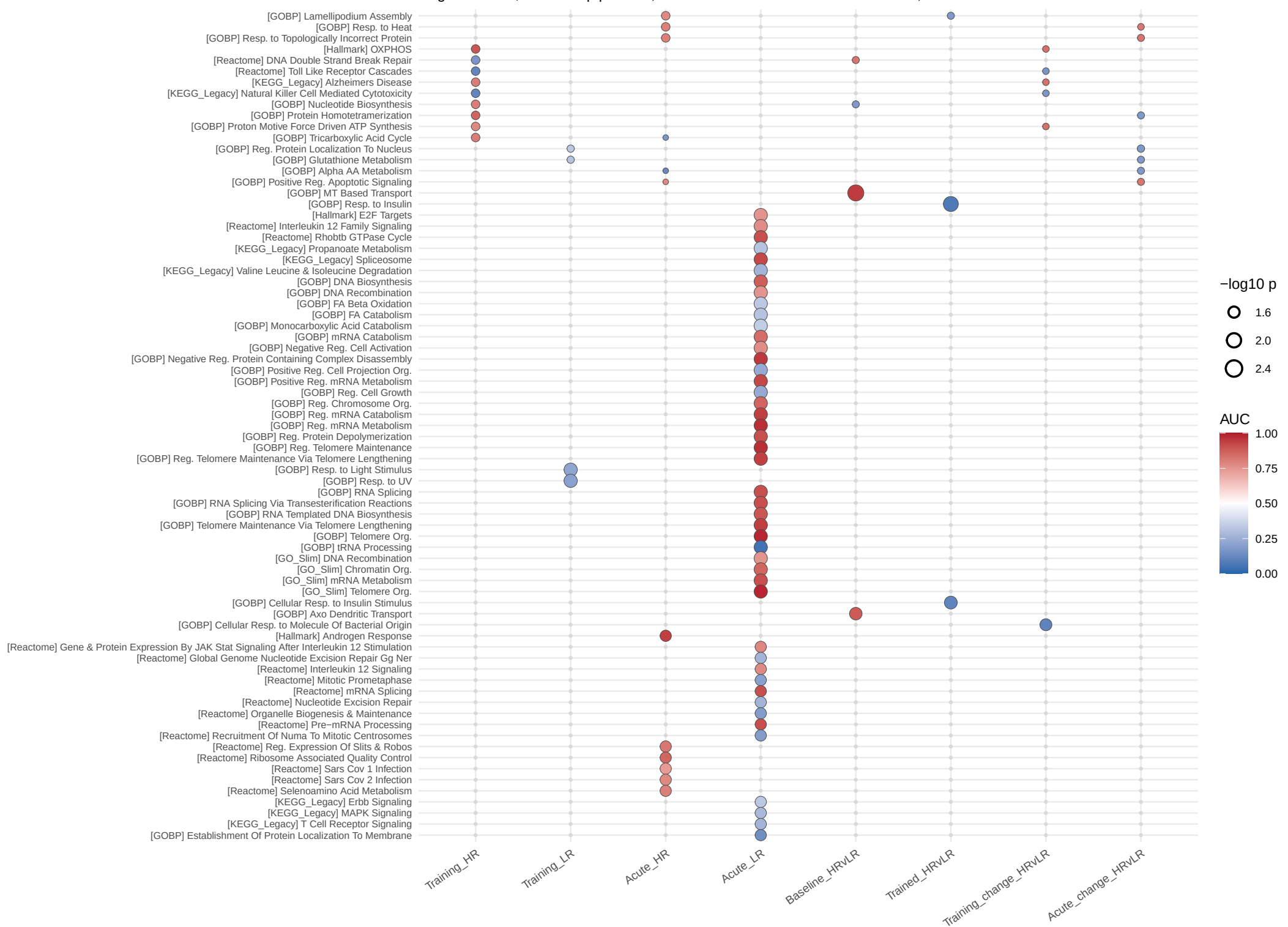
Set classification hits (1/5)

singscore AUC, Wilcoxon p per task; 362 sets nominal in at least one column; rows 1–75



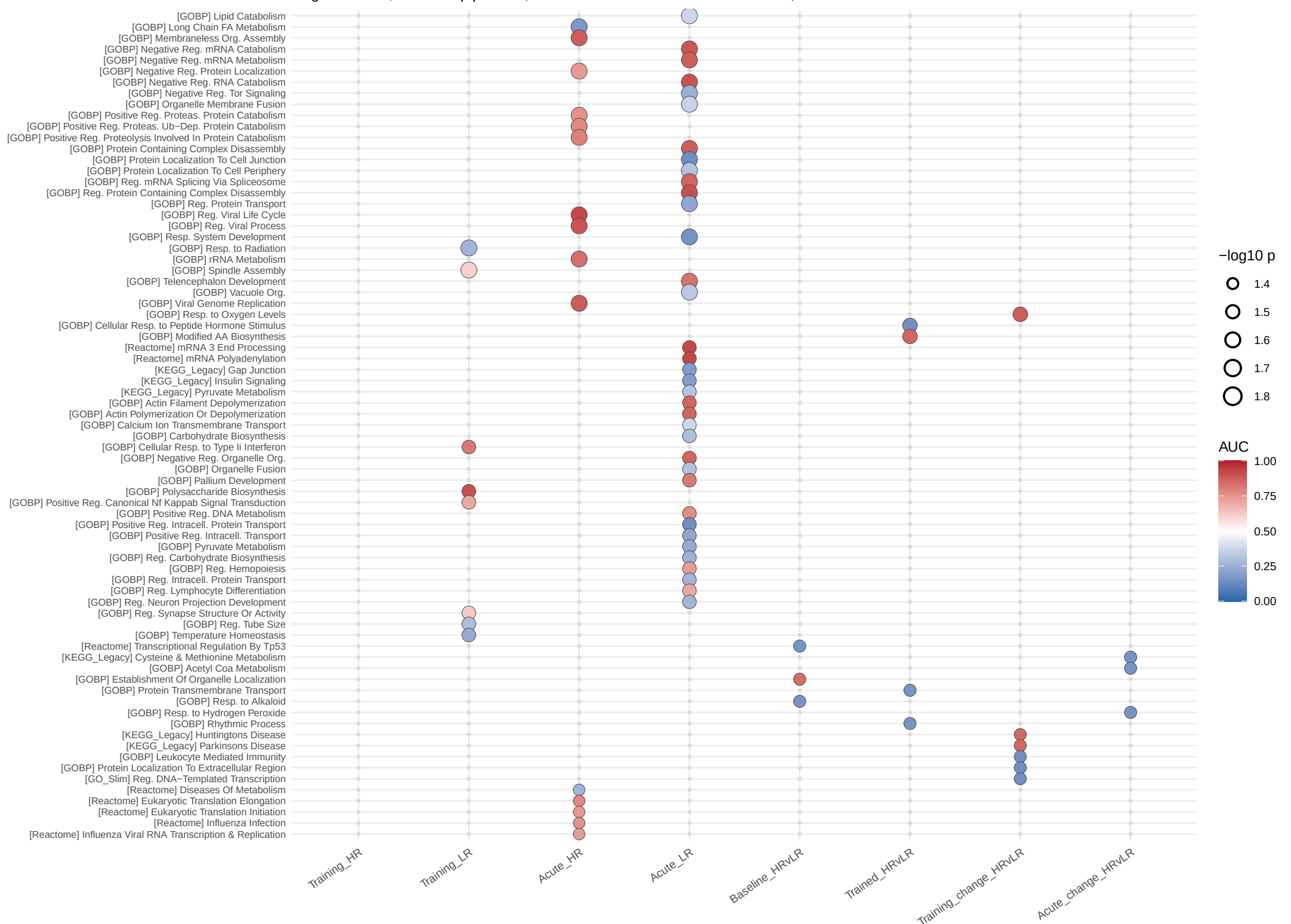
Set classification hits (2/5)

singscore AUC, Wilcoxon p per task; 362 sets nominal in at least one column; rows 76–150



Set classification hits (3/5)

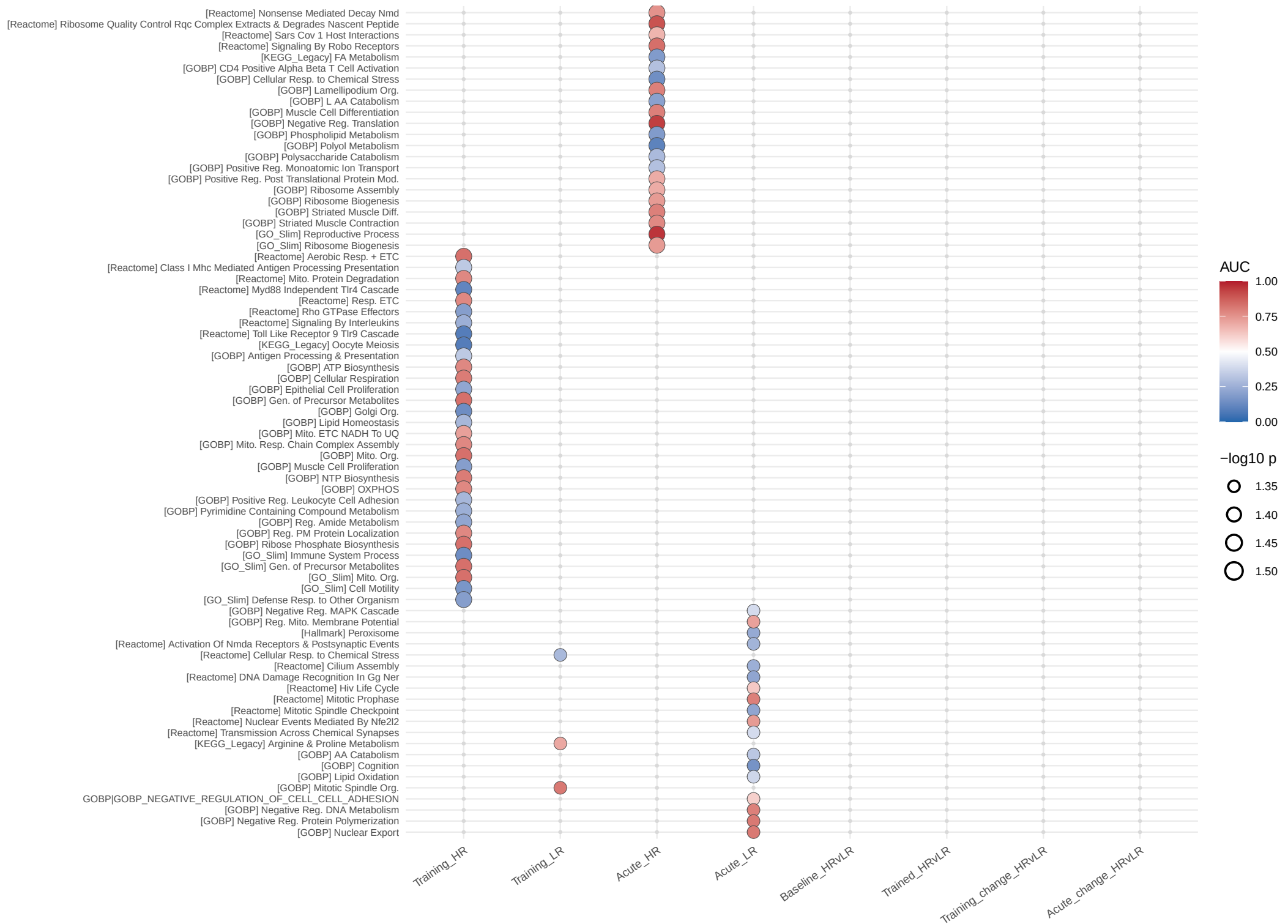
singscore AUC, Wilcoxon p per task; 362 sets nominal in at least one column; rows 151–225



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal, fill = AUC, size = $-\log_{10} p$. Black ring: BH < 0.05 within collection. Grey speck: not nominal

Set classification hits (4/5)

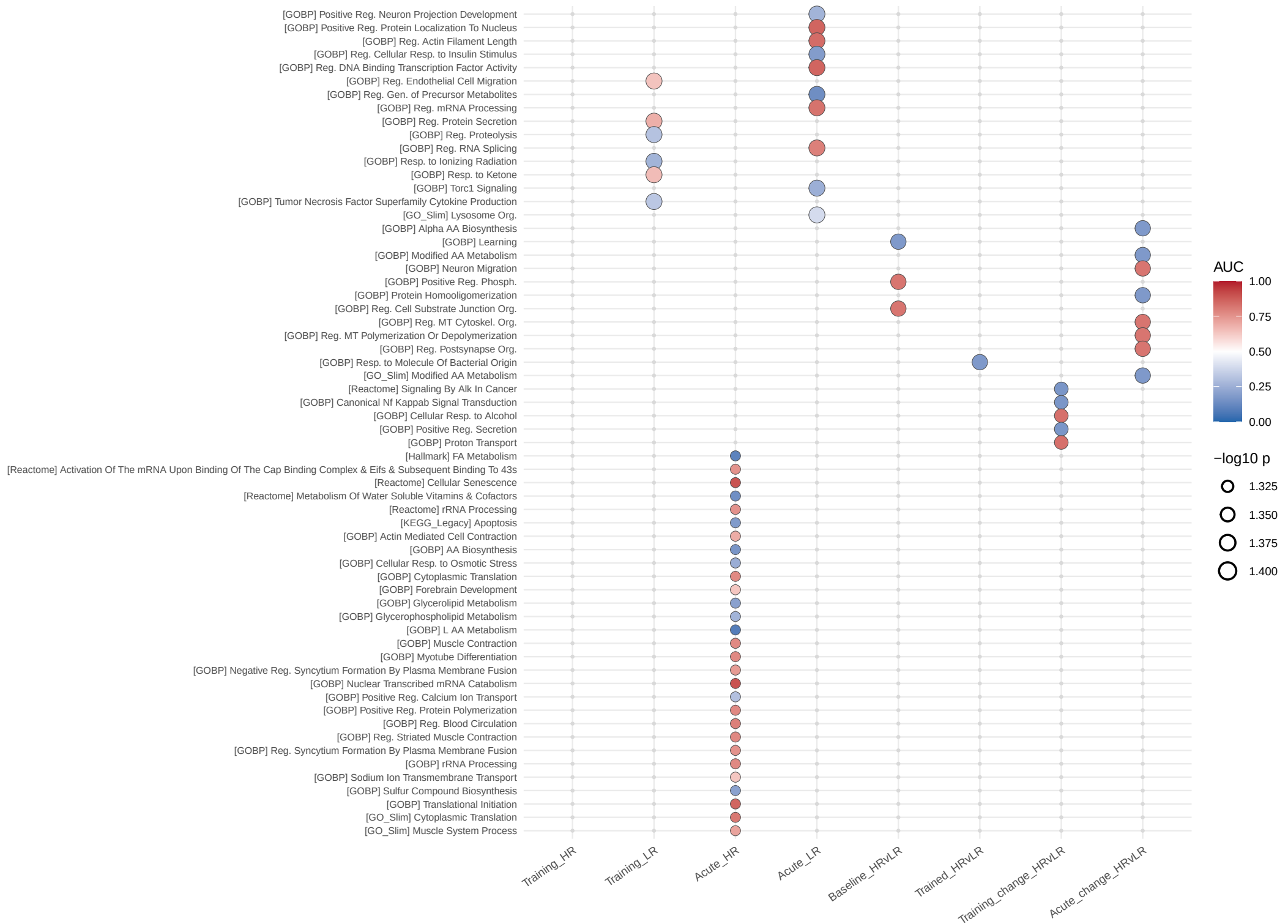
singscore AUC, Wilcoxon p per task; 362 sets nominal in at least one column; rows 226–300



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p. Filled dot: nominal, fill = AUC, size = $-\log_{10} p$. Black ring: BH < 0.05 within collection.

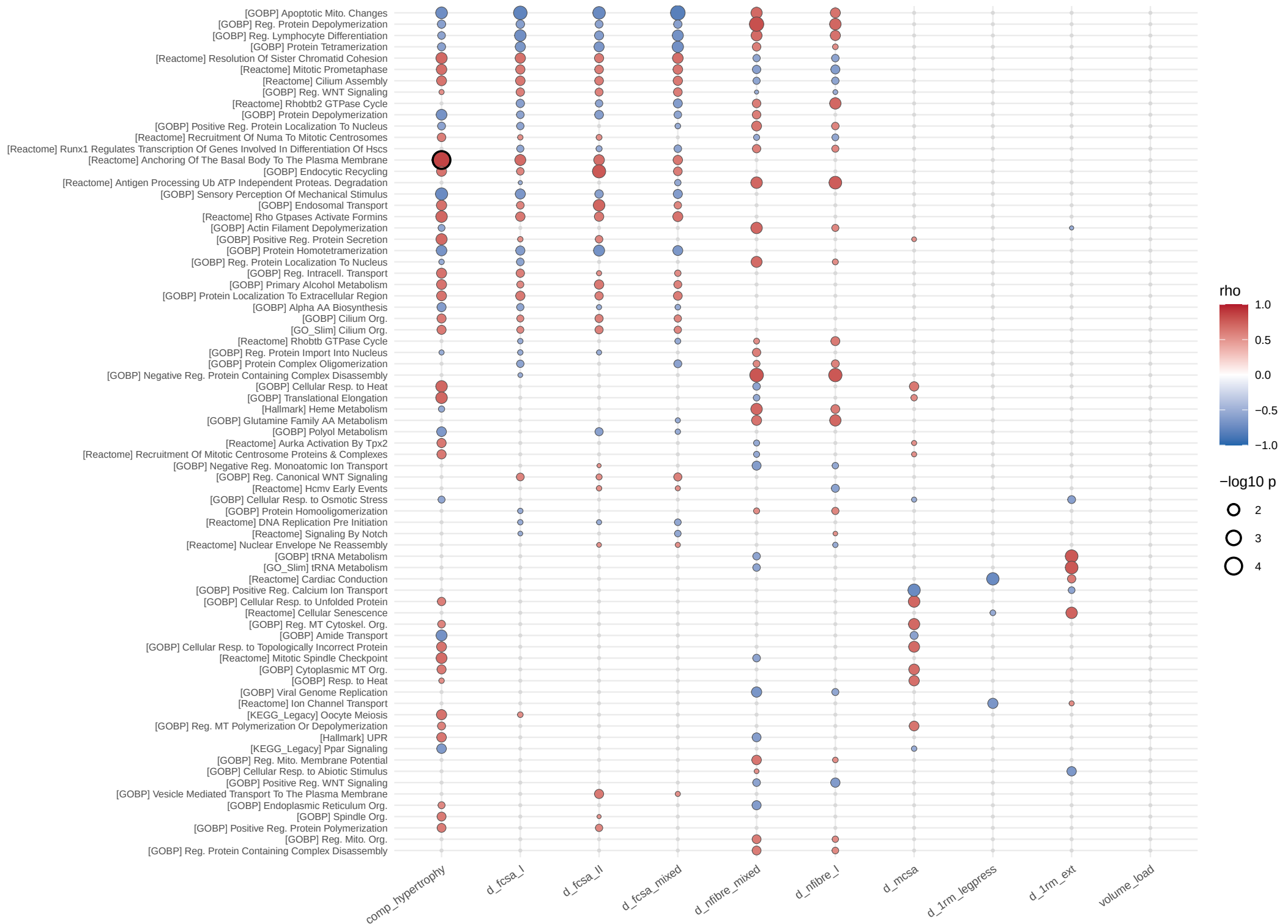
Set classification hits (5/5)

singscore AUC, Wilcoxon p per task; 362 sets nominal in at least one column; rows 301–362



Set association hits, acute change (T3 – T2) (1/5)

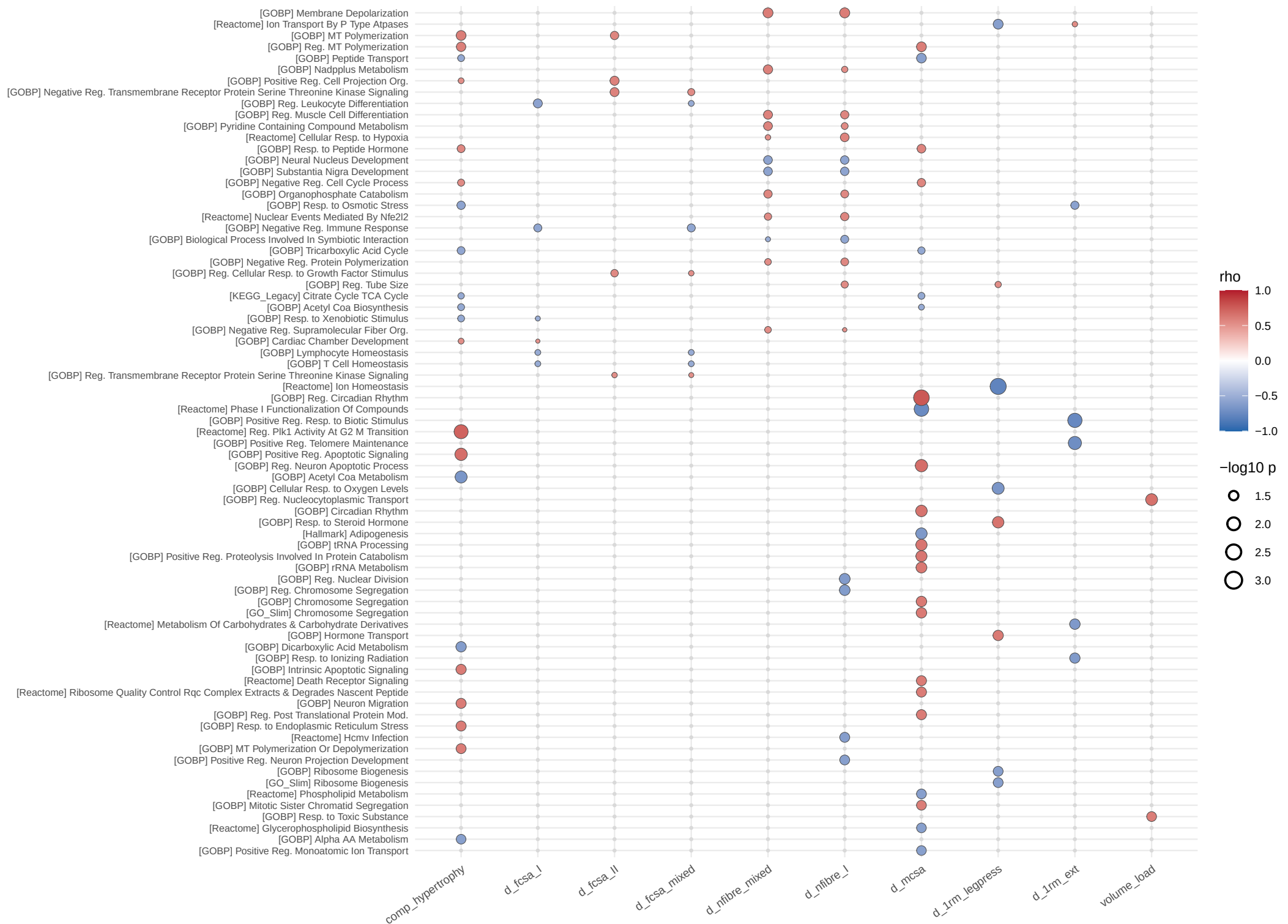
Spearman against each phenotype; 302 sets nominal in at least one column; rows 1–75



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log10 p. Black ring: BH < 0.05 within collection. Grey

Set association hits, acute change (T3 – T2) (2/5)

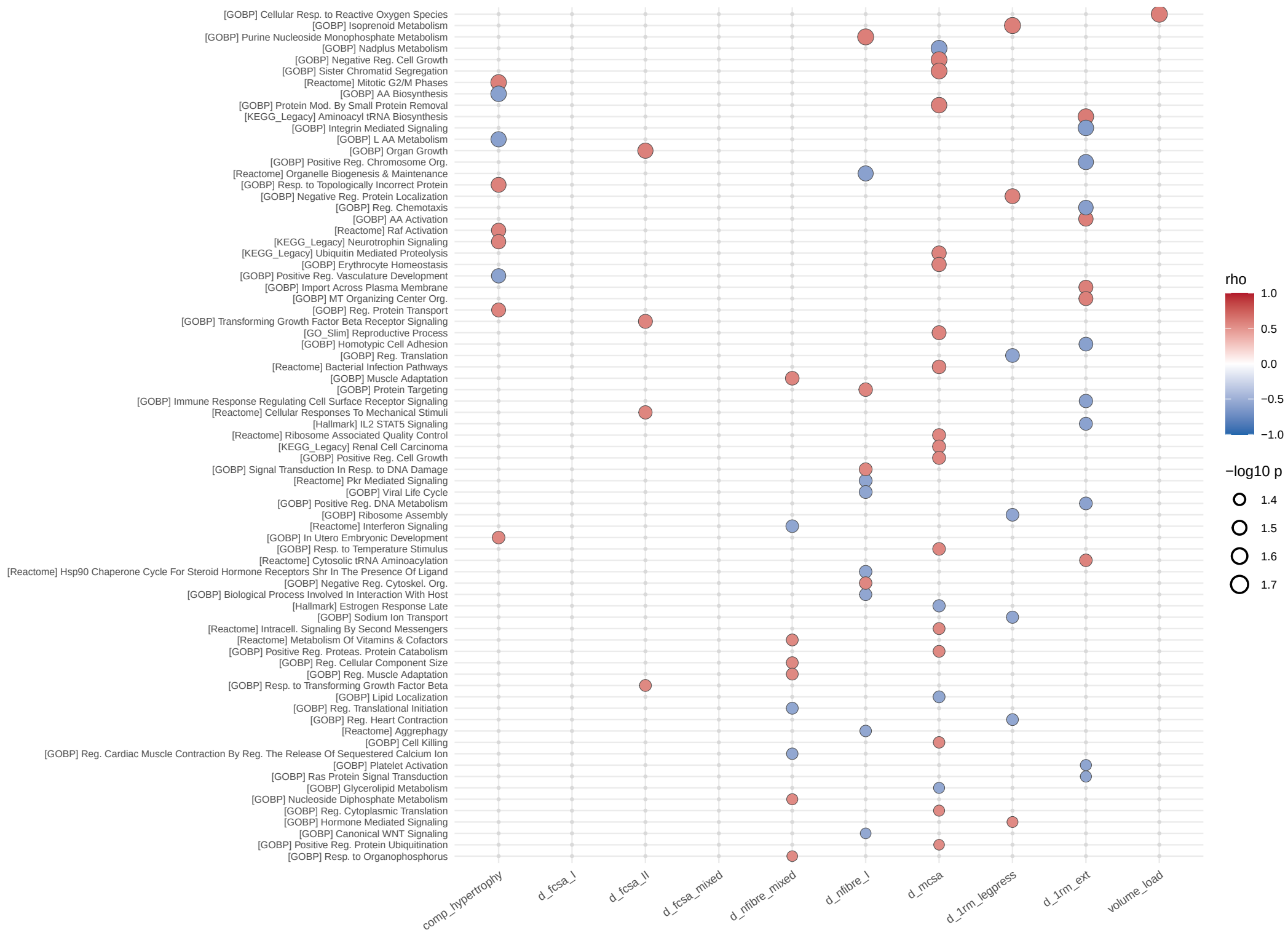
Spearman against each phenotype; 302 sets nominal in at least one column; rows 76–150



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH < 0.05 within collection.

Set association hits, acute change (T3 – T2) (3/5)

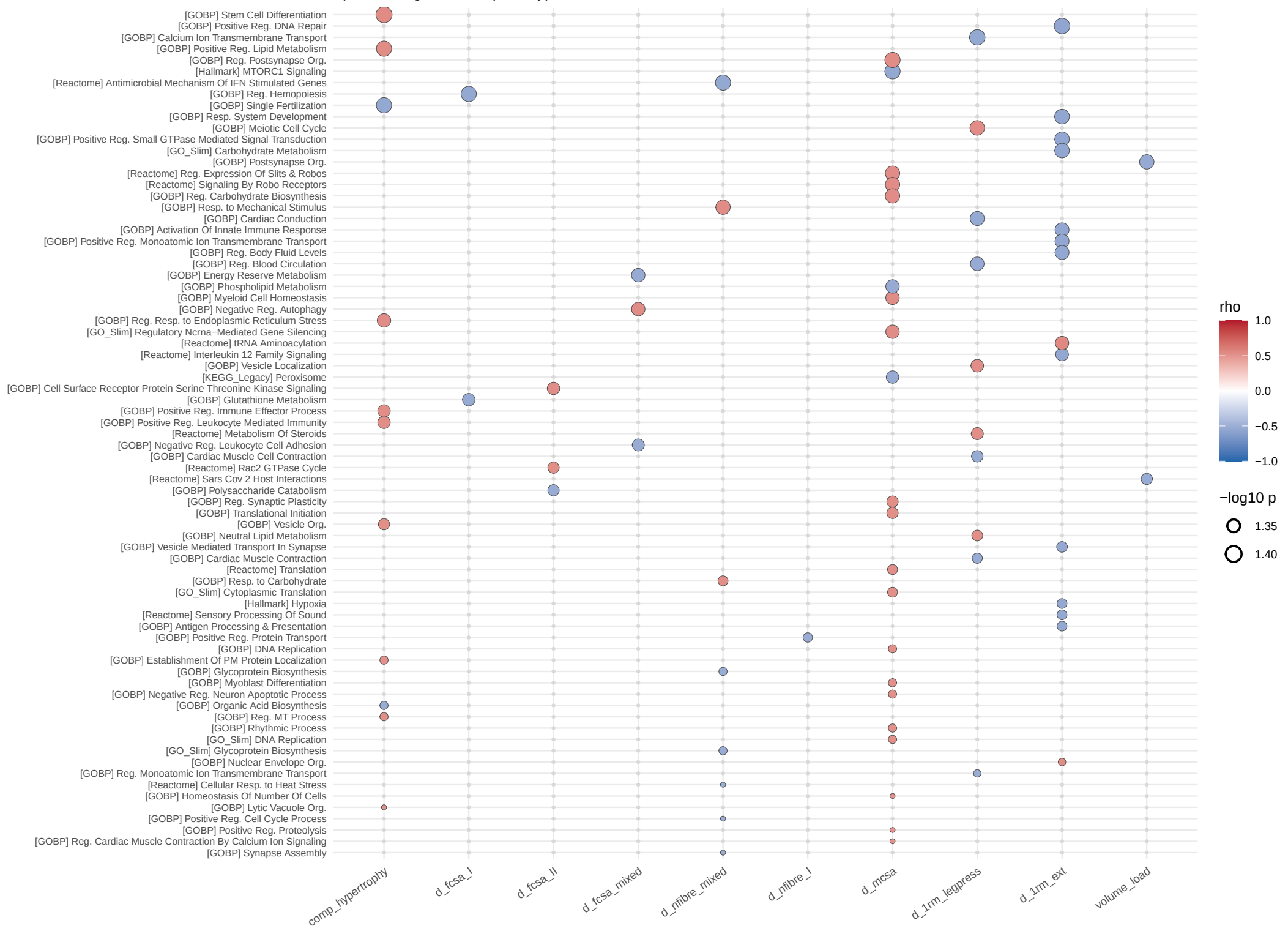
Spearman against each phenotype; 302 sets nominal in at least one column; rows 151–225



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH < 0.05 within column

Set association hits, acute change (T3 – T2) (4/5)

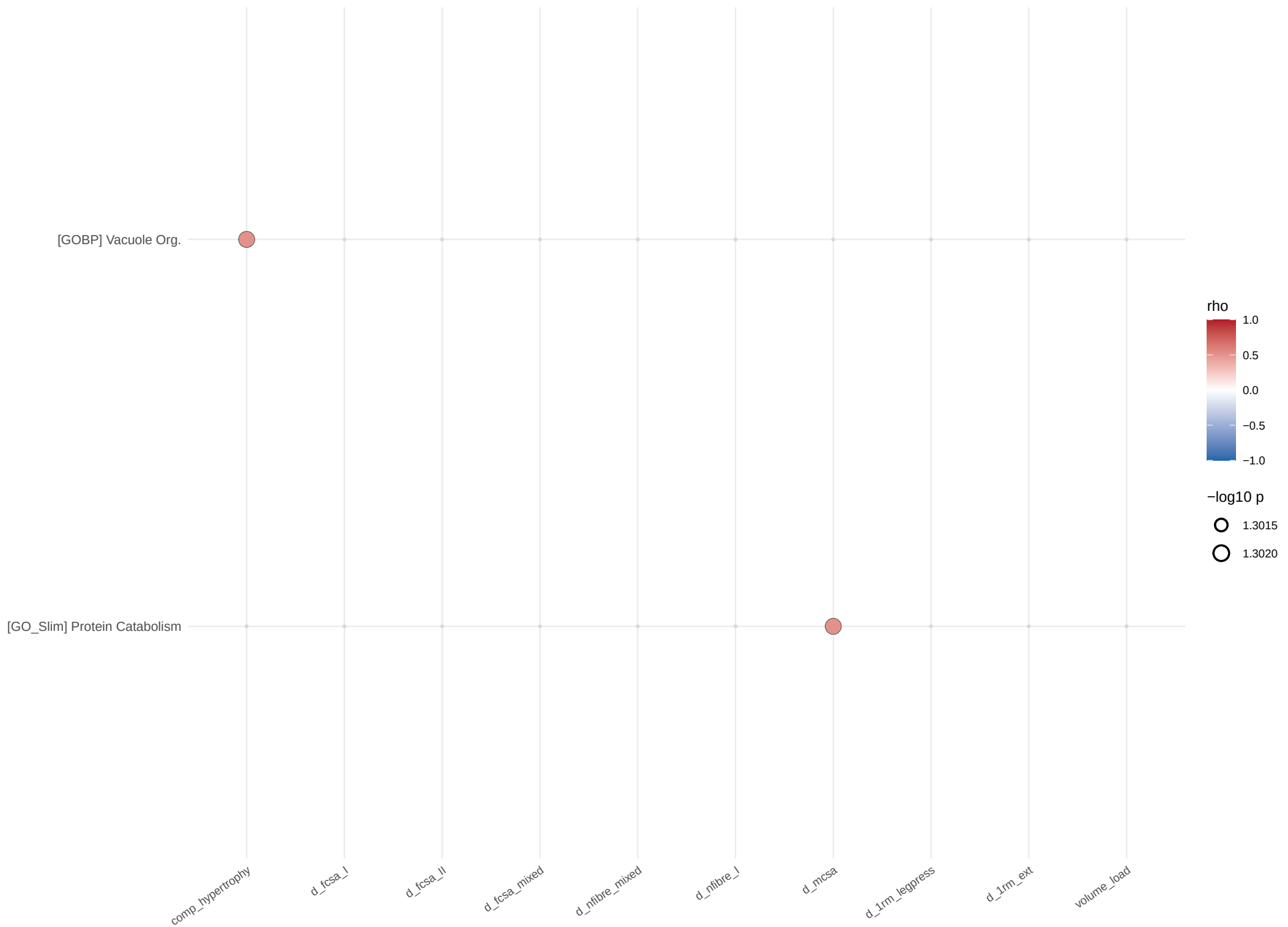
Spearman against each phenotype; 302 sets nominal in at least one column; rows 226–300



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log₁₀ p. Black ring: BH < 0.05 within collection. Grey speck: not

Set association hits, acute change (T3 – T2) (5/5)

Spearman against each phenotype; 302 sets nominal in at least one column; rows 301–302



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH < 0.05 within collection. Grey speck: not nominal. Table: c_data/05_c

Set association hits, level at T1 (1/7)

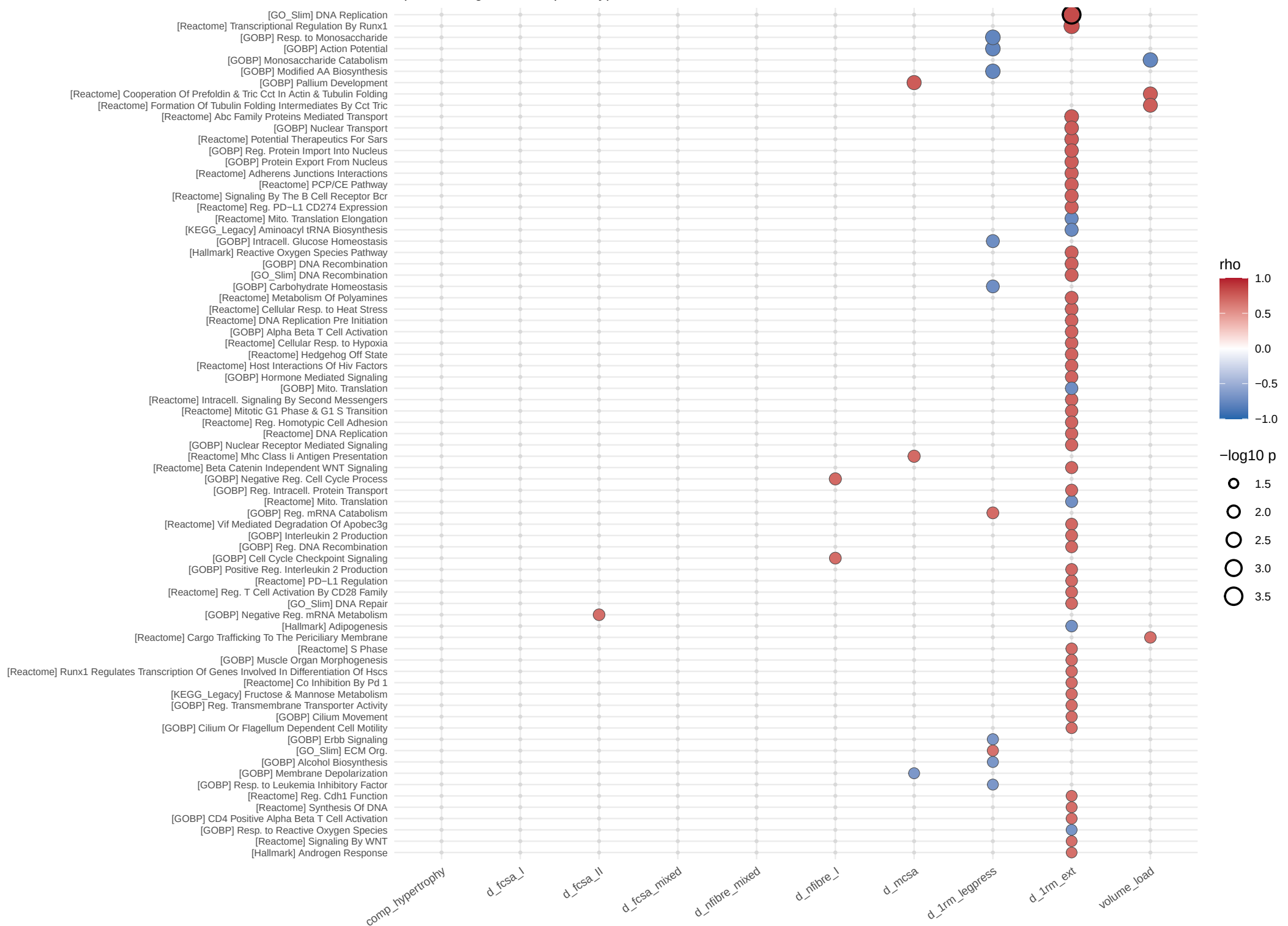
Spearman against each phenotype; 470 sets nominal in at least one column; rows 1–75



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log₁₀ p. Black ring: BH < 0.05 within column

Set association hits, level at T1 (2/7)

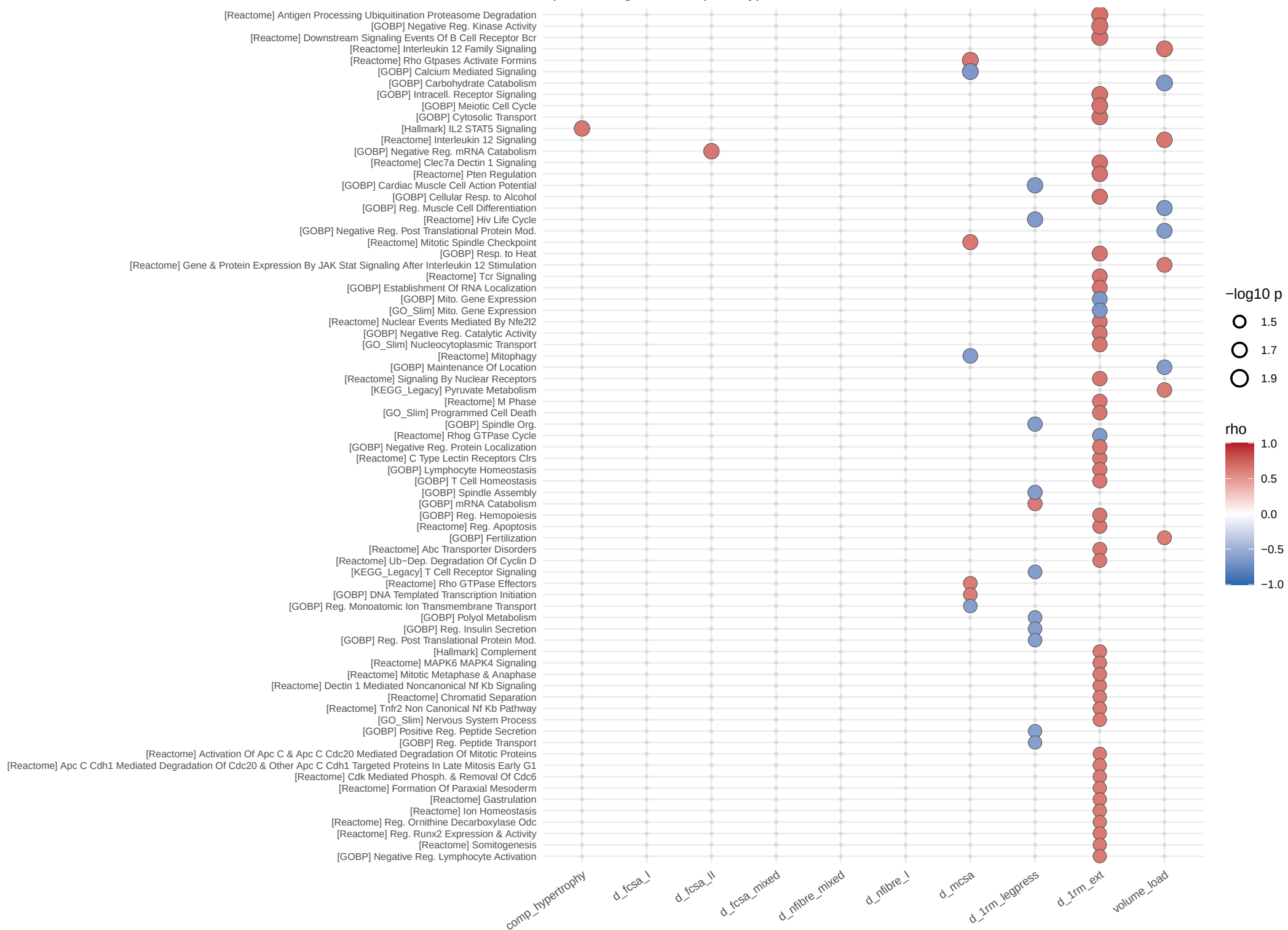
Spearman against each phenotype; 470 sets nominal in at least one column; rows 76–150



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH < 0.05 within collection. Grey

Set association hits, level at T1 (3/7)

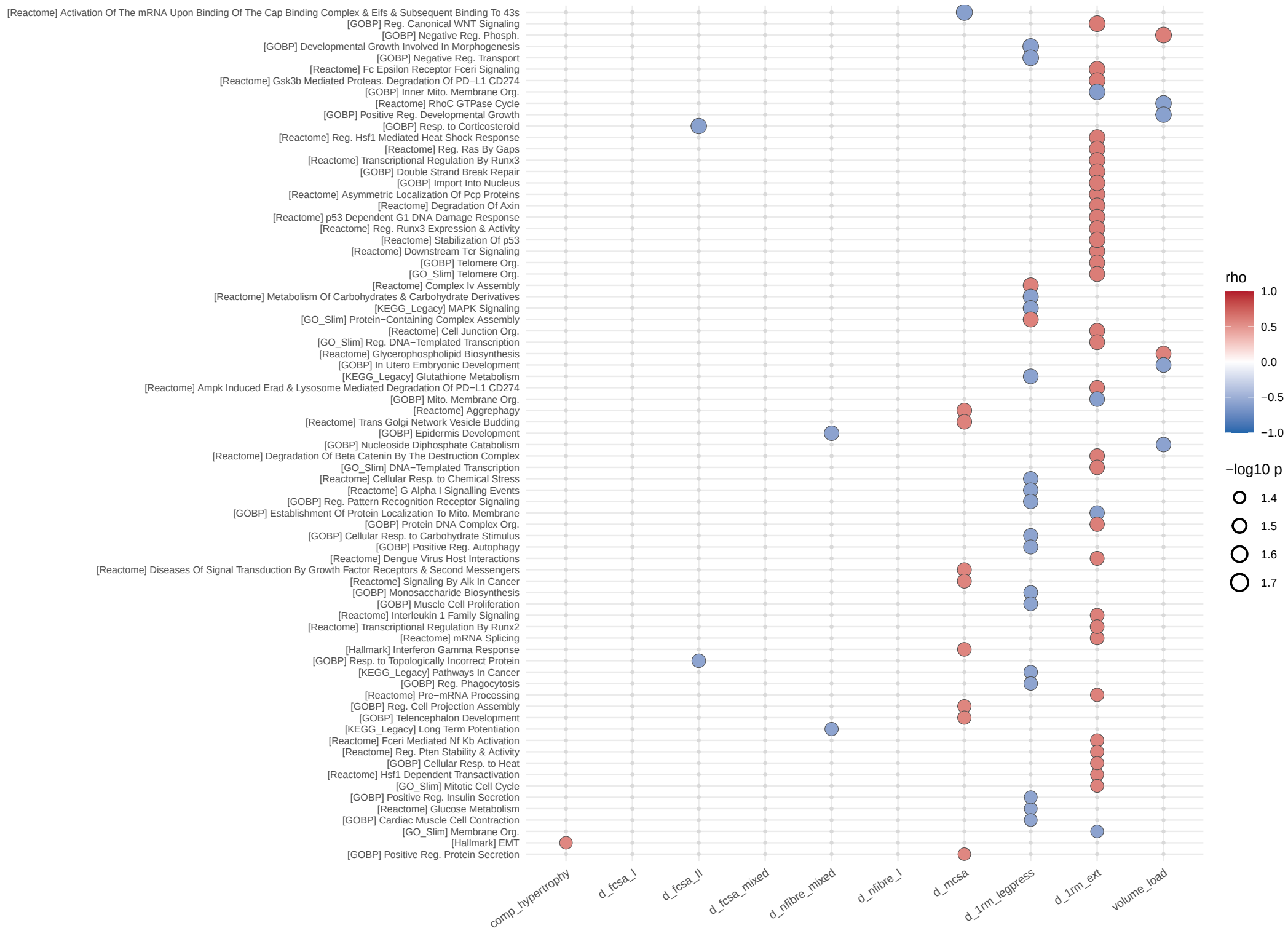
Spearman against each phenotype; 470 sets nominal in at least one column; rows 151–225



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH

Set association hits, level at T1 (4/7)

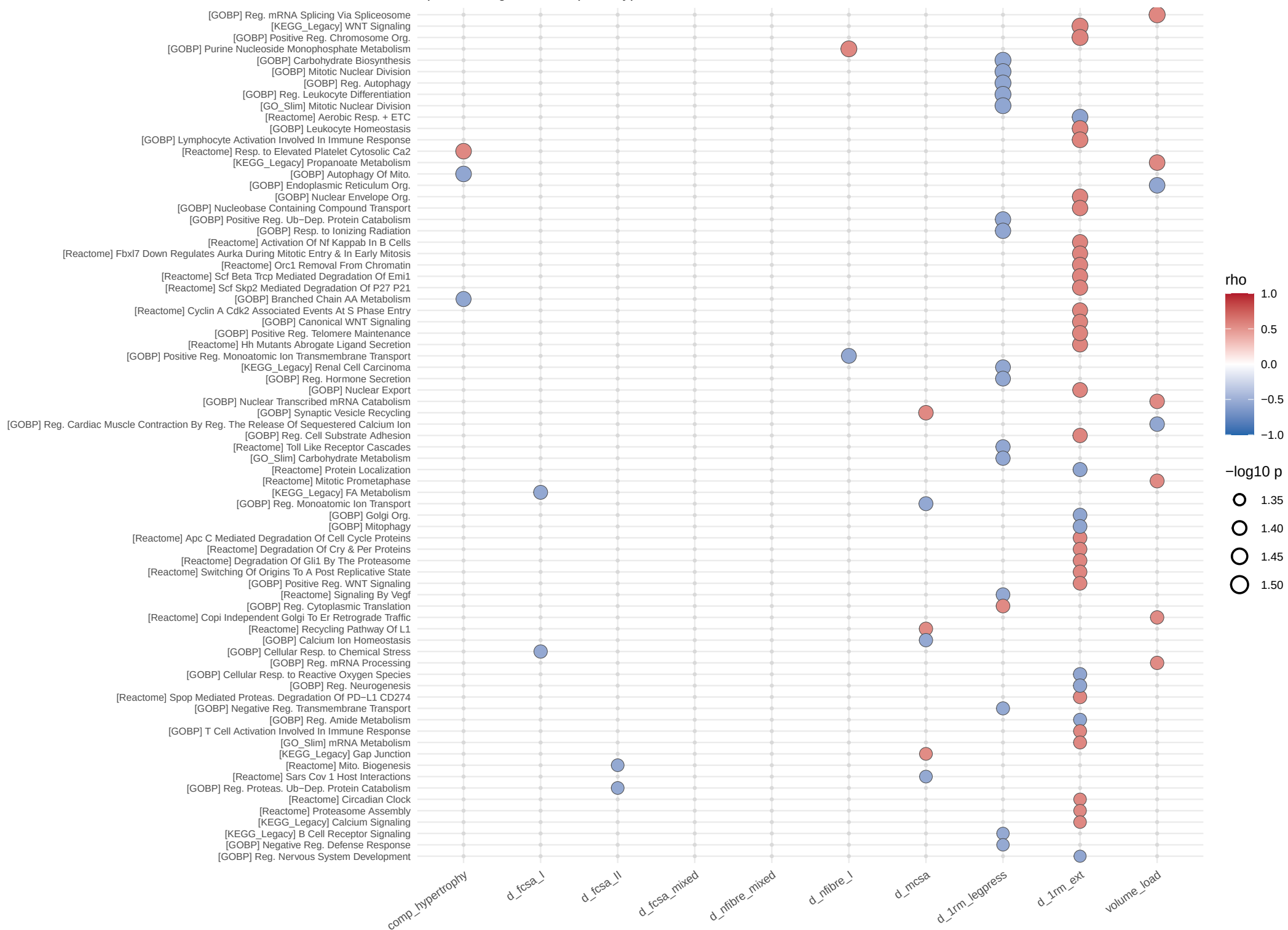
Spearman against each phenotype; 470 sets nominal in at least one column; rows 226–300



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH $q < 0.05$.

Set association hits, level at T1 (5/7)

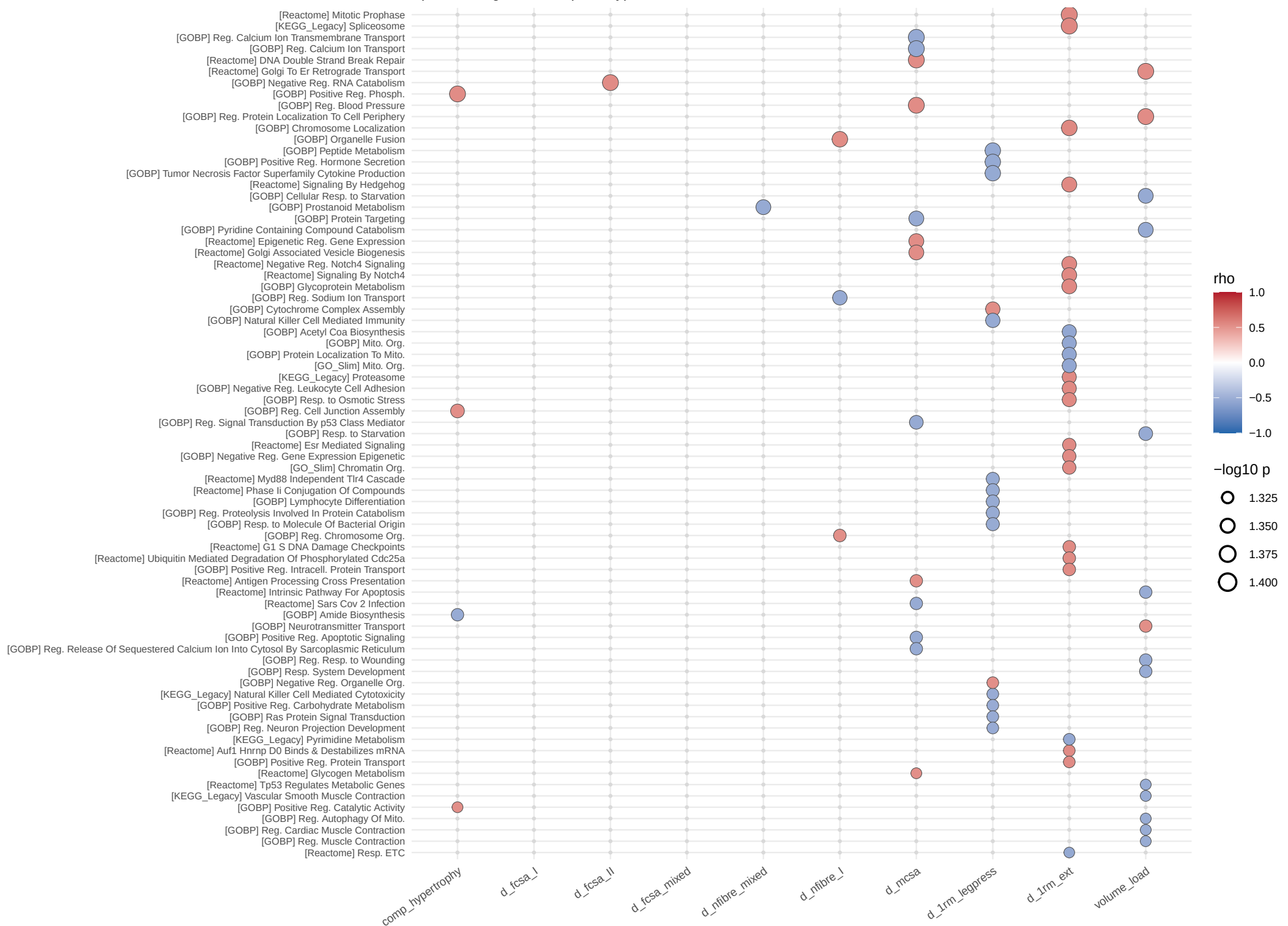
Spearman against each phenotype; 470 sets nominal in at least one column; rows 301–375



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log10 p. Black ring: BH < 0.05 within collection.

Set association hits, level at T1 (6/7)

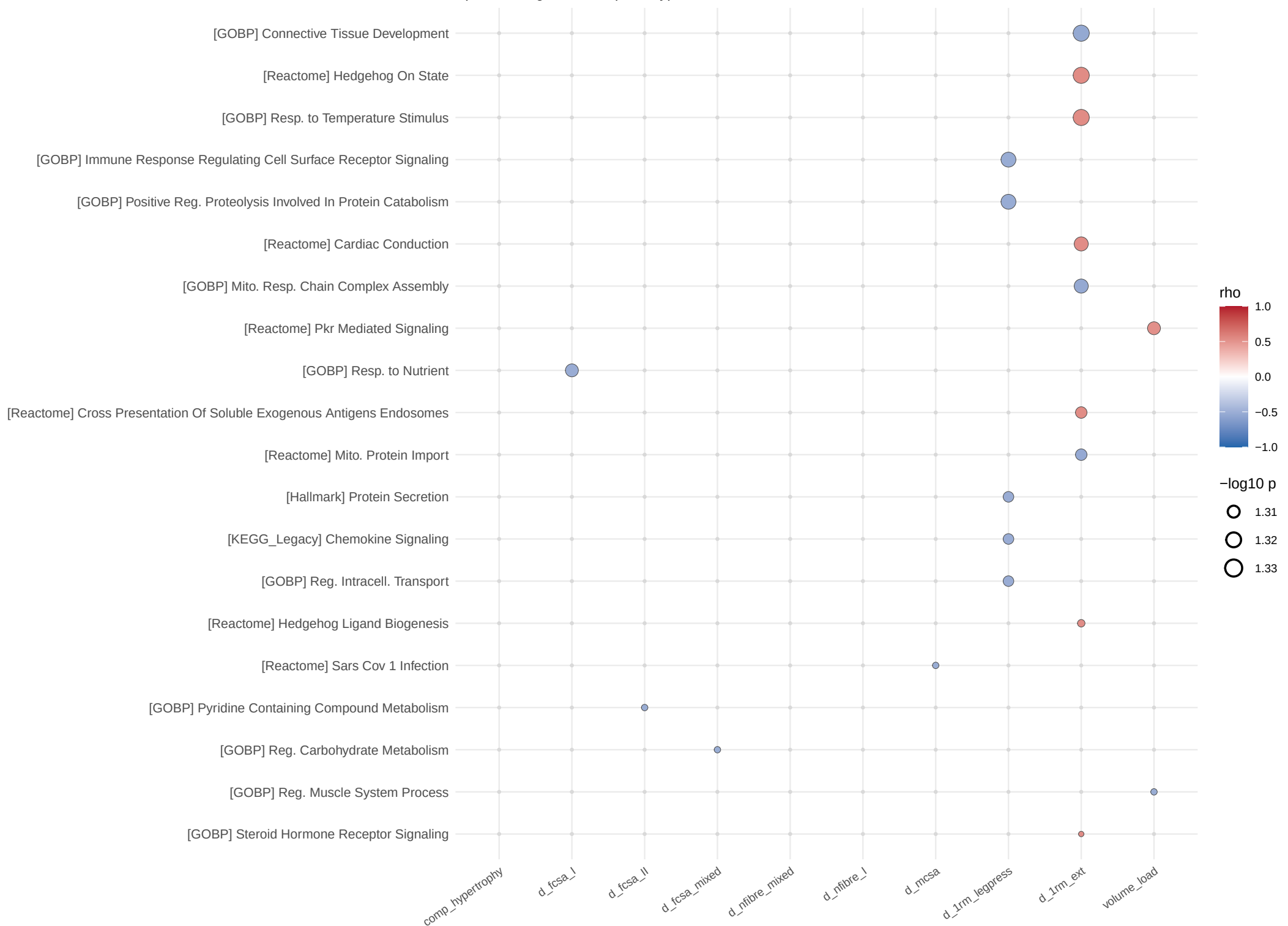
Spearman against each phenotype; 470 sets nominal in at least one column; rows 376–450



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log10 p. Black ring: BH < 0.05 within collection.

Set association hits, level at T1 (7/7)

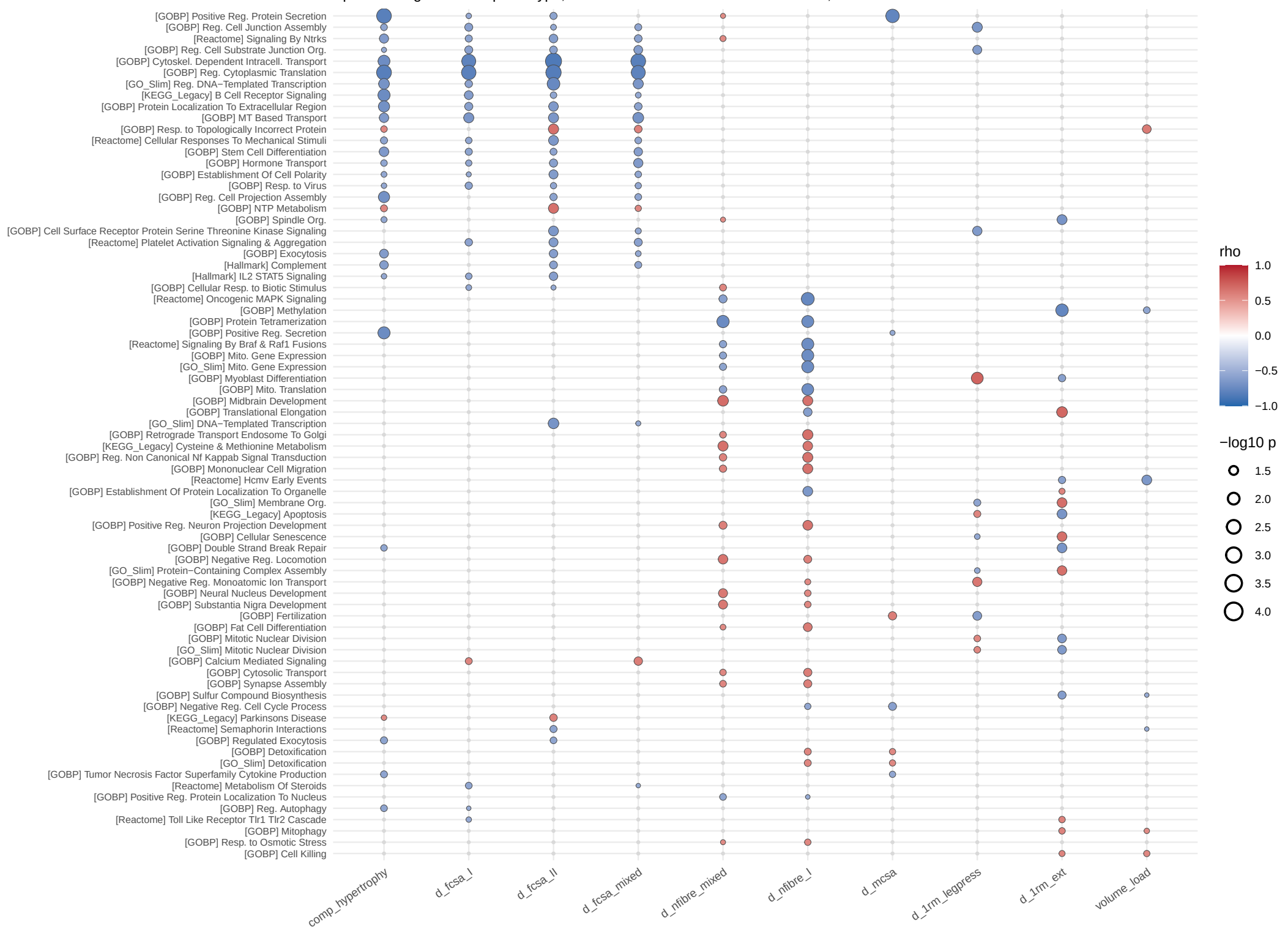
Spearman against each phenotype; 470 sets nominal in at least one column; rows 451–470



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log10 p. Black ring: BH < 0.05 within column

Set association hits, training change (T2 - T1) (1/6)

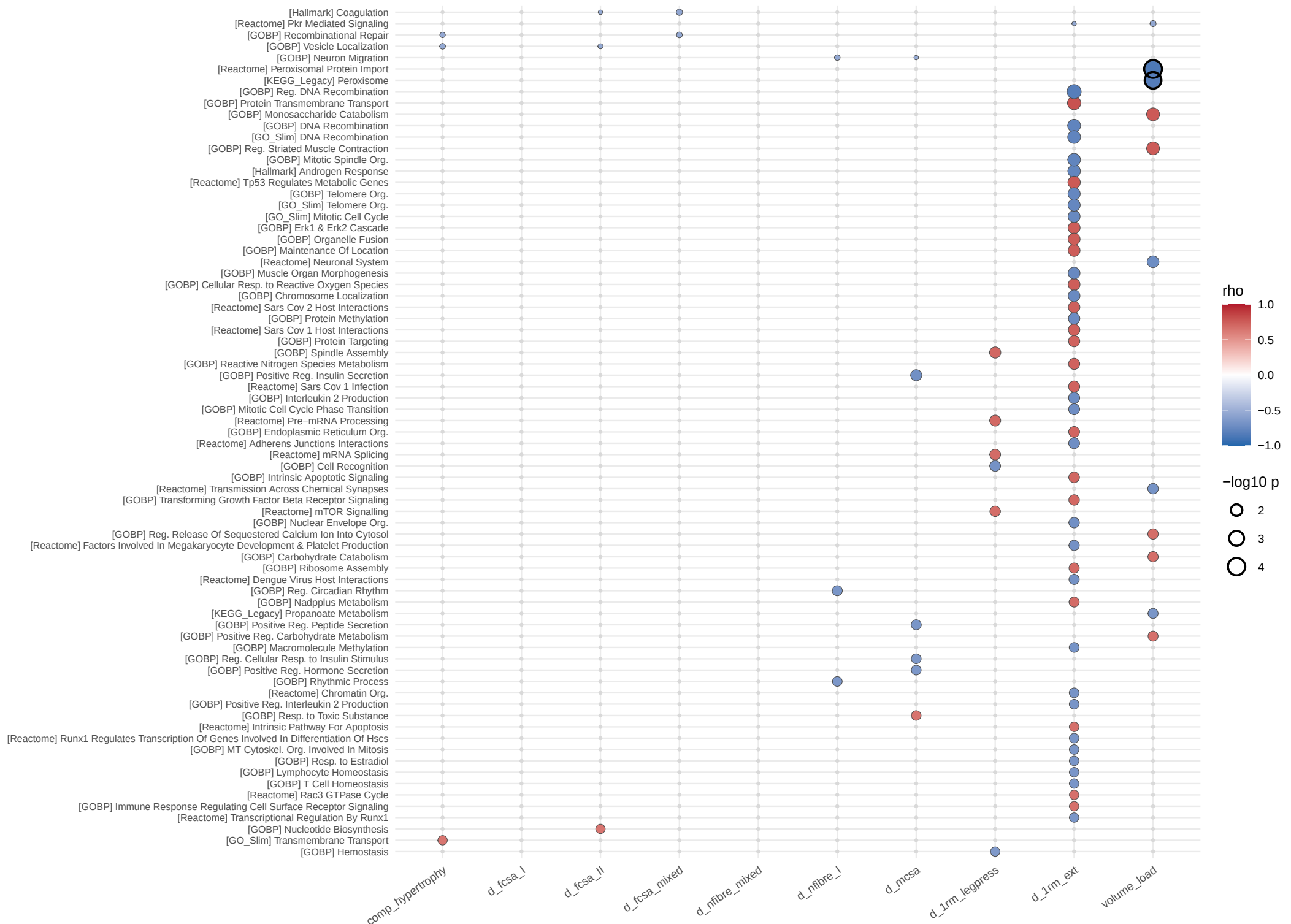
Spearman against each phenotype; 393 sets nominal in at least one column; rows 1-75



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH < 0.05 within collection. Grey speck: not

Set association hits, training change (T2 – T1) (2/6)

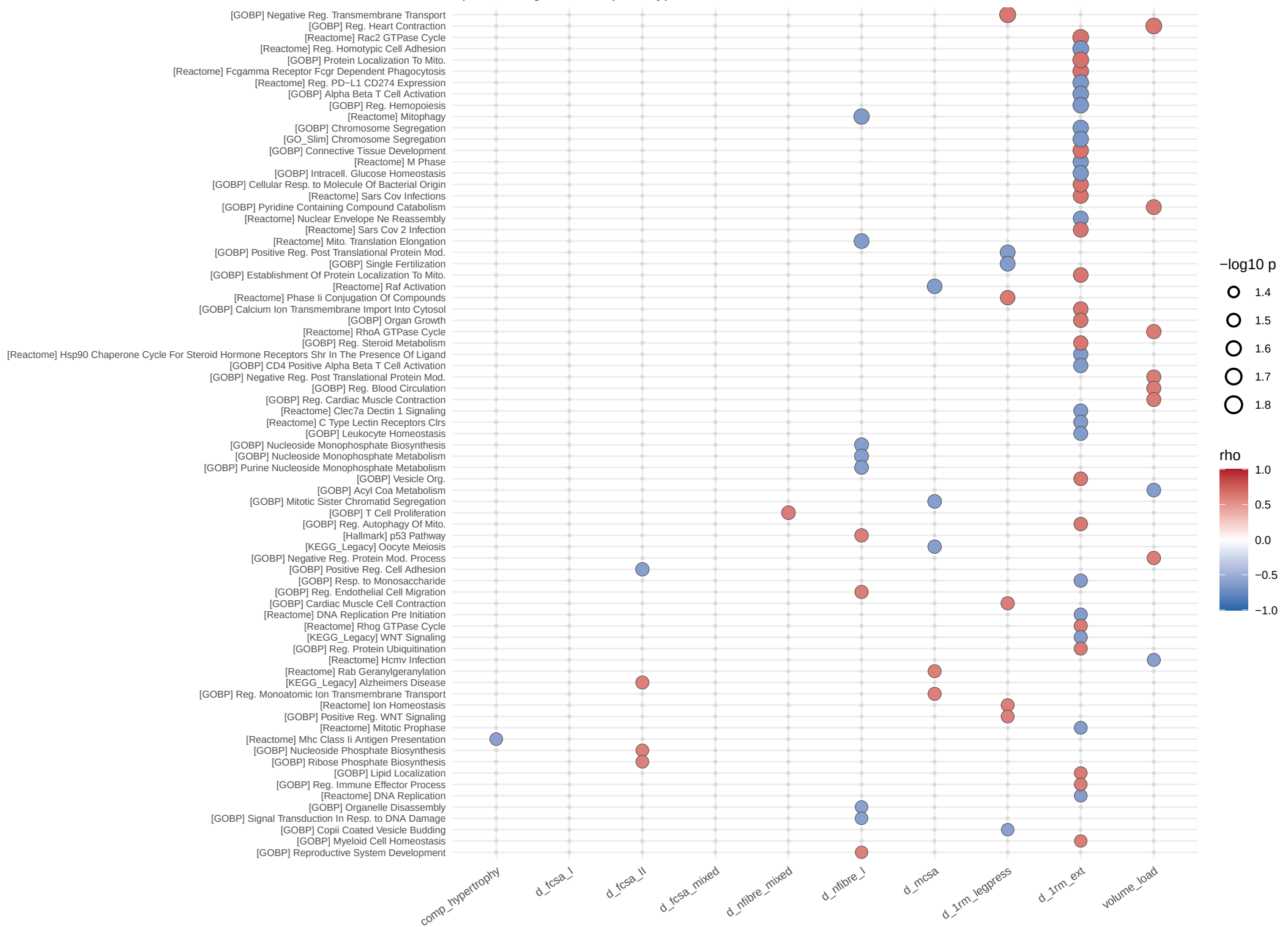
Spearman against each phenotype; 393 sets nominal in at least one column; rows 76–150



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log10 p. Black ring: BH < 0.05 within collection. Grey

Set association hits, training change (T2 – T1) (3/6)

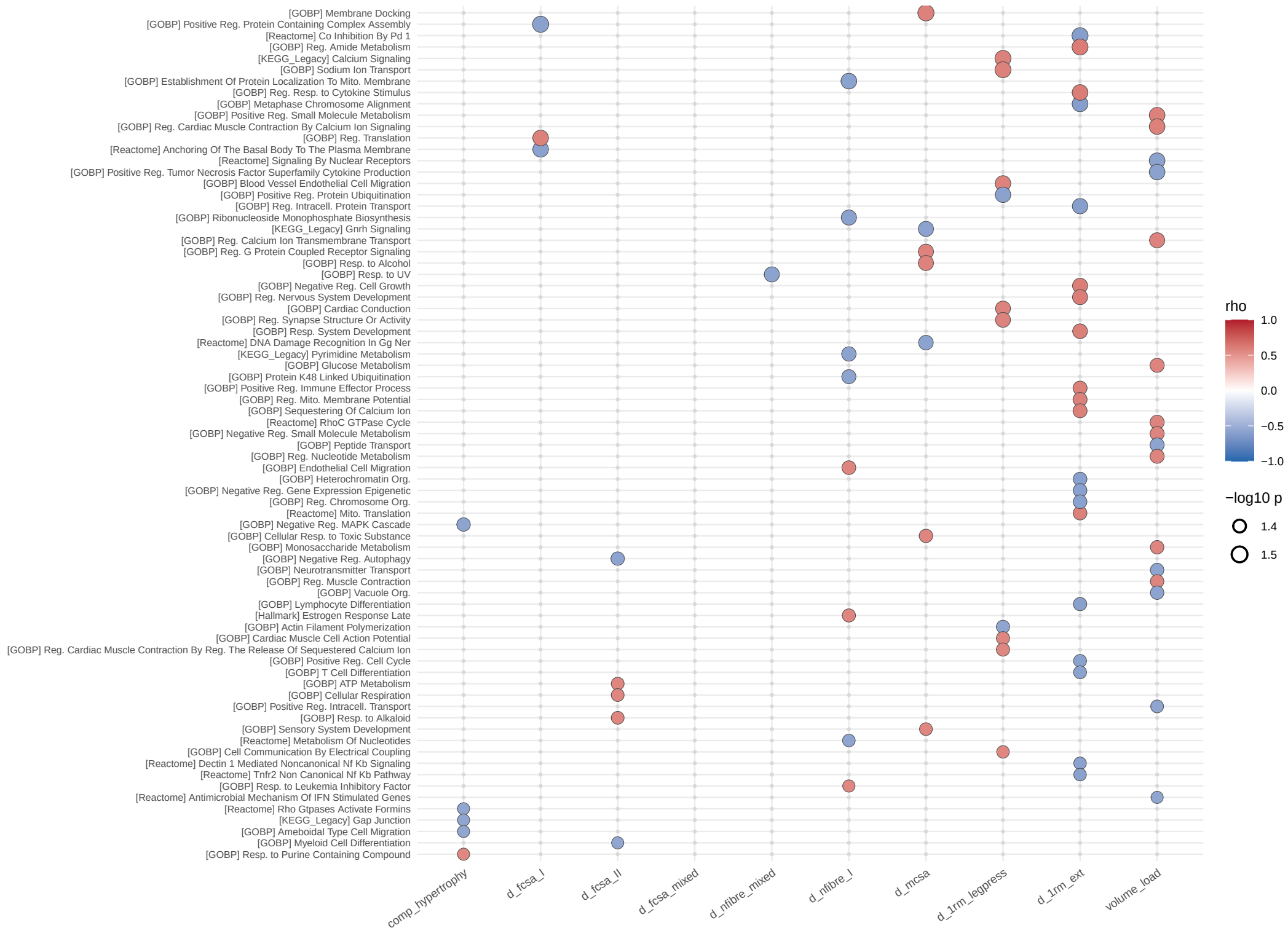
Spearman against each phenotype; 393 sets nominal in at least one column; rows 151–225



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = $-\log_{10} p$. Black ring: BH < 0.05 within column

Set association hits, training change (T2 – T1) (4/6)

Spearman against each phenotype; 393 sets nominal in at least one column; rows 226–300



Rows: sets at nominal p < 0.05 in at least one column, most columns first, then by best p. Filled dot: nominal, fill = rho, size = -log10 p. Black ring: BH < 0.05 within collection.

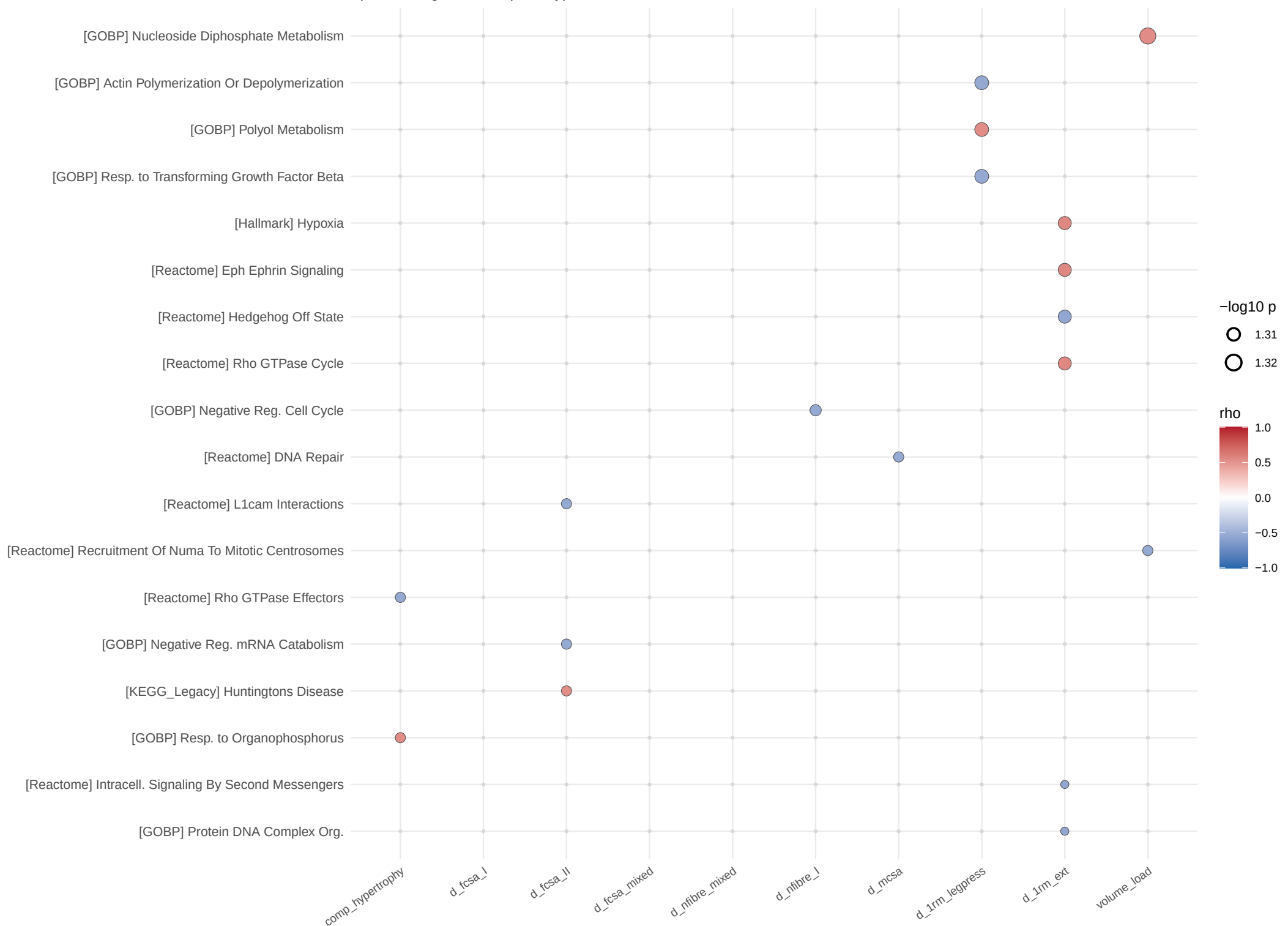
Set association hits, training change (T2 – T1) (5/6)

Spearman against each phenotype; 393 sets nominal in at least one column; rows 301–375



Set association hits, training change (T2 – T1) (6/6)

Spearman against each phenotype; 393 sets nominal in at least one column; rows 376–393



Rows: sets at nominal $p < 0.05$ in at least one column, most columns first, then by best p . Filled dot: nominal, fill = ρ , size = $-\log_{10} p$. Black ring: BH < 0.05 within collection. Grey speck: